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FINAL YEAR DISSERTATION PROJECT

SAFE STEP - THE PROACTIVE JOURNEY COMPANION

CST3990 UNDERGRADUATE INDIVIDUAL PROJECT

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Abstract

The UK’s growing elderly population faces disabilities that preclude their independence, take a toll on caregivers, and cost the UK government millions per year. This project presents the design and development of a mobile and web application that assists cognitively and physically impaired users during their TfL journeys. Built as a cross-platform solution on React Native and FastAPI, SafeStep employs cutting-edge predictive technology from Fountech AI to analyse the user’s profile and foresee potential journey failure scenarios and subsequently deliver proactive actions aimed at keeping users safe. SafeStep went through a live testing period in which users took the application with them on their TfL journeys and evaluated several key features against their assigned personal profile. Testing results show great potential in improving users' confidence when solo travelling while providing valuable feedback for future work.

1. Introduction

The United Kingdom's population was estimated to be 69.3 million in 2024 (Office for National Statistics, 2025), with 19% of people belonging to the 65+ age group (UK Parliament, 2024). This age group percentage is expected to encompass over 25% of the population by 2045, and continue to rise thereafter. The present project recognises the important health issues of frailty and dementia within this age group, and develops an application that, through the use of predictive AI technologies, aims to prolong the independence of these individuals, support their carers, and, in doing so, reduce costs for the NHS.

The developed application, SafeStep, identifies and tackles in-journey risks leveraging Anticip8, a predictive model provided by FountechAI. Given an appropriate context, Anticip8 is capable of predicting the next most probable action. Through providing the user's detailed profile, SafeStep promptly identifies multiple potential journey failure points and subsequently generates useful interventions that the app can perform, ensuring the user reaches their destination safely. Accurate interventions that efficiently prevent journey risks from taking place can restore independent travel, alleviating stress from both the user and caregiver. SafeStep is available in both a mobile working prototype, designed for users to take with them through their journeys, and a web interface, presenting a visualisation tool to examine trips' failure risks, system-generated interventions, and their perceived impact.

This report is divided into 5 chapters, including an introduction, literature review, product overview, testing, and conclusion. The literature review chapter analyses the current state of the problem the elderly population faces, the impact on their health, and the burden this has on the UK government and institutions. Additionally, it introduces current technologies aimed at solving some of these issues and their blind spots or areas of improvement. The product overview chapter details the development and implementation of SafeStep's core features, architecture and user interface. The testing chapter shows the analysis and performance results used to determine the best model combination to generate failure nodes and probability calculations. Additionally, this chapter discusses the insights gathered through the live testing phase, where real users were asked to test SafeStep's journey planning capabilities. Finally, the conclusion engages in a brief discussion on the project overview, testing phase results and acknowledges project limitations and future work.

2. Literature Review

2.1 Physical and Cognitive Vulnerability in the Elderly Population

2.1.1 Frailty and Its Functional Implications

Frailty is defined as a state of vulnerability resulting from the natural decline in homeostatic reserves and the functional capacity of major organ systems (Clegg et al., 2013; Pomatto et al., 2017). Further significant characteristics of frailty include the progressive loss of muscle strength and the subsequent lifestyle and health changes caused by stressors such as falling, infections, and confusion (Clegg et al., 2013).

According to the British Geriatrics Society (2025), frailty is a common condition that affects over 10% of the 65+ demographic and over 50% of the 85+ demographic. Research has found that frailty costs UK healthcare systems £5.8 billion per year. This number is driven by increased hospital admissions, longer inpatient stays, and increased general practice consultations (Han et al., 2019).

Age UK (2020) states that people living with frailty have to adapt how they manoeuvre through their daily life and complete previously trivial tasks. This adaptation is the hardest when the frail population leave their homes. While there can be a large number of home adaptations to adapt to each individual’s needs, public spaces are the most unforgiving. The outdoors has an enormous impact on people’s quality of life, and the ability to safely navigate this environment is key to maintaining independence (Inclusive Design for Getting Outdoors, 2007). Poorly maintained pavements and a lack of supportive facilities, amongst others, are the main barriers that prevent the elderly population from having a healthy relationship with their immediate neighbourhood. Furthermore, environmental issues and how they affect public infrastructure also hold a significant weight on the elderly population’s independence. Chiang et al. (2024) found that lower temperatures are an important factor for bone fractures in older adults, with wet and icy surfaces being a major source of falls. Additionally, they concluded that colder temperatures may result in impaired muscle function, joint stiffness and reduced reaction times, all significant contributing factors that heighten fall risk.

Even when the issue is not environmentally or infrastructurally related, there is an additional risk residing in a frail individual's psyche. Joung (2008) highlights the psychological toll the fear of falling has on older adults. Previous fall history is a significant contributor to developing a fear of falling, with individuals who have fallen multiple times or sustained significant injuries naturally being more prone to being afraid of falling. Furthermore, activity restriction and isolation are common issues that directly affect the independence of elderly individuals, preventing them from utilising available infrastructure, even when it's technically accessible.

2.1.2 Dementia and Its Cognitive Repercussions

Dementia is a group of related symptoms that affect key areas of the brain involved in the successful completion of daily tasks (NHS, 2024). Alzheimer's disease is a type of dementia that makes up a significant amount of cases (NHS, 2024). Alzheimer's disease affects the brain by inhibiting the communication mechanisms between neurons, eventually resulting in them perishing(National Institute of Ageing, 2024). Specific areas of the brain that are affected early on are the hippocampus, which has a major role in the encoding of working memory, significantly influencing spatial memory (Toniolo et al., 2025; Eichenbaum, 2017). Acute atrophy in this area contributes to the inability to maintain a mental map of one's surroundings, crucial to independent navigation. Furthermore, the frontal lobe is an additional compromised target, having an important function in planning, language and social interactions, and general executive function (Alzheimer's Society, 2018). These are critical skills to be relied on during the independent completion of normal daily tasks and activities.

According to the UK's Alzheimer's Society (2024b), by 2024, there were an estimated 982,000 people living with dementia, with an estimated over a third of the people suffering from the condition being undiagnosed. At the time, the cost of dementia for the UK was forecasted to be over £42 billion. By 2040, the number of people with dementia is expected to grow to 1.4 million, signifying a cost of over £90 billion. Furthermore, while unpaid carers supporting someone affected by dementia save the UK economy a considerable amount of money (Alzheimer's Society, 2024c), devoting their time to doing so limits their ability to be productive members of society.

In a metropolitan city like London, with a significantly developed and complex public transport system, urban mobility is a task that requires memory and spatial navigation skills. The impairment of key areas of the brain allowing effective navigation is one of the earliest cognitive decline symptoms indicating oncoming dementia

(Coughlan et al., 2018). This means even familiar routes can result in disorientation and distress. Hauger et al. (2019) explored some of the issues navigating public transport poses for elderly people with dementia. Amongst other remarks, they found that many wrongly estimate the passing of time. This has a significant impact on the use of public transport since a common occurrence is immediately leaving a stop after their arrival, or completely miscalculating the intended arrival time altogether. For example, this could be arriving at a bus stop hours before or after their bus is due.

Communication and social behaviour are major components in outdoor life and travelling through public transport systems. For those with dementia, informal rules and procedures such as coherent articulation or buying a ticket may be forgotten or significantly harder to perform (Hauger et al., 2019). These issues only contribute to the individual’s fragile state of mind. According to the Alzheimer's Society (2024a), the general anxiety and panic caused by stressful situations, such as getting lost or wandering, can make symptoms of dementia worsen. Furthermore, getting lost or its assumed risk is a leading cause of permanent care home residency (McShane et al., 1998).

2.2 Institutionalisation

While the NHS and local authorities might cover the costs of permanent care home residency, many who are not as fortunate can find this financially unfeasible. Reports state that over 50% of care home residents are affected by frailty (British Geriatrics Society, 2025), and over 70% are affected by dementia (Alzheimer's Society, 2025). The combination of cognitive issues associated with dementia and the physical and psychological vulnerability caused by frailty raises the need to trade the individual’s independence for the perceived safety of a care home. However, research has found that institutionalisation plays a significant role in cognitive decline (Harsányiová et al., 2018). While rooted in several issues, significant causes include a lack of social and physical activity, impoverished environments and sustained psychological stress. This makes for the development of proactive technologies aimed at mitigating potential risks, a key component for prolonging independent living.

2.3 Current System Issues

2.3.1 NHS as a Reactive System

In his 2024 independent investigation of the NHS, Darzi (2024) highlights the reactive nature of how the NHS currently operates. He argues that the system is in a "critical condition", allocating its resources and attention to how to react to and treat illness rather than investing in preventive measures that primarily aim at keeping patients healthier for longer. Additionally, he places significant weight on how paramount it is to move care into communities while reinforcing the use of new and existing technologies. This idea is rooted in what is described as the "left shift" strategy, which focuses on moving care closer to home on the basis that hospital-centred systems are economically unviable and logistically inefficient for a population living with multiple chronic illnesses. In shifting the concentrated traffic from hospitals to community-based care, the system can mitigate a state where individuals overflow and worsen the waiting time in acute facilities.

2.3.2 The Technological Gap

While current navigational applications like Google Maps, Citymapper or TfL GO are extremely popular and have revolutionised urban mobility for the general population, they do not account for the cognitive demands of the elderly. This is particularly true for those who suffer from the mild cognitive impairment characteristic of early-stage dementia (Knopman et al., 2014). The primary way they fail the elderly population is in how they suffer from the same "reactive" nature that Darzi (2024) claims the NHS does. Mistakes, such as taking a wrong turn or an early disembark, are communicated to the user after they happen, overrelying on their ability to recover from mistakes without getting overwhelmed. This delay between the misdirection and the correction puts a tremendous cognitive burden on the individual to self-correct in an already sensory-overwhelming environment.

Furthermore, these applications provide an overly informative user interface designed for tech-savvy users. The constant real-time updates, multiple route variations and maps with several interest points do not account for how easily overwhelmed a distressed person can be. Dementia has an acute impact on how the brain handles new information. The hippocampus, being a crucial part of the brain tasked with filtering irrelevant information in the encoding and maintenance stage of working memory, is an area that is particularly targeted by Dementia (Toniolo et al., 2025). For a person with cognitive decline, the inability to filter out incoming information (whether that's from loud bystanders or phone's push notifications) while trying to follow navigational instructions correctly, can lead to a dangerous sensory overload.

While proactive applications helping users navigate through convoluted public transport systems already exist, they use this approach to provide travel solutions to the general public or other vulnerable demographics. Unfortunately, most of these applications, such as WayMap for the visually impaired, or Moovit with features to warn users of crowded trains and buses, fail to adapt to the cognitive overload issues caused by traditional travel applications. Rapid state changes or constant, fast voice commands would likely be an addition to the numerous contributing factors that could trigger a panic or anxiety attack. On the other hand, applications that are targeted to the elderly tend to focus on other necessary aspects of their life that are afflicted by the impairments associated with ageing. Platforms like CarePredict or CareTaker.ai are either dependent on additional previous infrastructure, such as sensor systems or aim to solve different issues, such as predicting UTI infections.

As a result, we can observe that standard navigational applications fail to account for the elderly user's potentially vulnerable state. They presuppose a calm, rational user, capable of interpreting complex data in a tumultuous situation. They force elderly users to adapt to the model user they create, rather than the applications adapting to their needs.

2.4 Foundational Models

2.4.1 Introduction to Foundational Models

Before 2017, the state-of-the-art technology regarding machine learning models for processing sequential data was Recurrent Neural Networks (RNNs) and their variants (e.g. Long Short-Term Memory and Gated Recurrent Units). These models efficiently processed sequential data and maintained context over time, but through a linear approach. This means that, for the model to understand the tenth element in a sequence, it first needs to process the previous nine elements. This presents a problem when it is faced with significantly large input sequences, as it results in long training times and the strains associated with long-range dependencies, such as trying to relate information processed at a very early step (Hochreiter & Schmidhuber, 1997; Hochreiter et al., 2001).

As an attempt to improve these areas, Vaswani et al. (2017) introduced the transformer architecture. This new neural network architecture employs a self-attention mechanism that allows the neural network to learn context based on the relationship of the sequential input, ingesting its elements simultaneously. This parallel processing is what makes transformers a more suitable alternative when presented with large, complex, and multimodal data (IBM, 2025d; Vaswani et al., 2017).

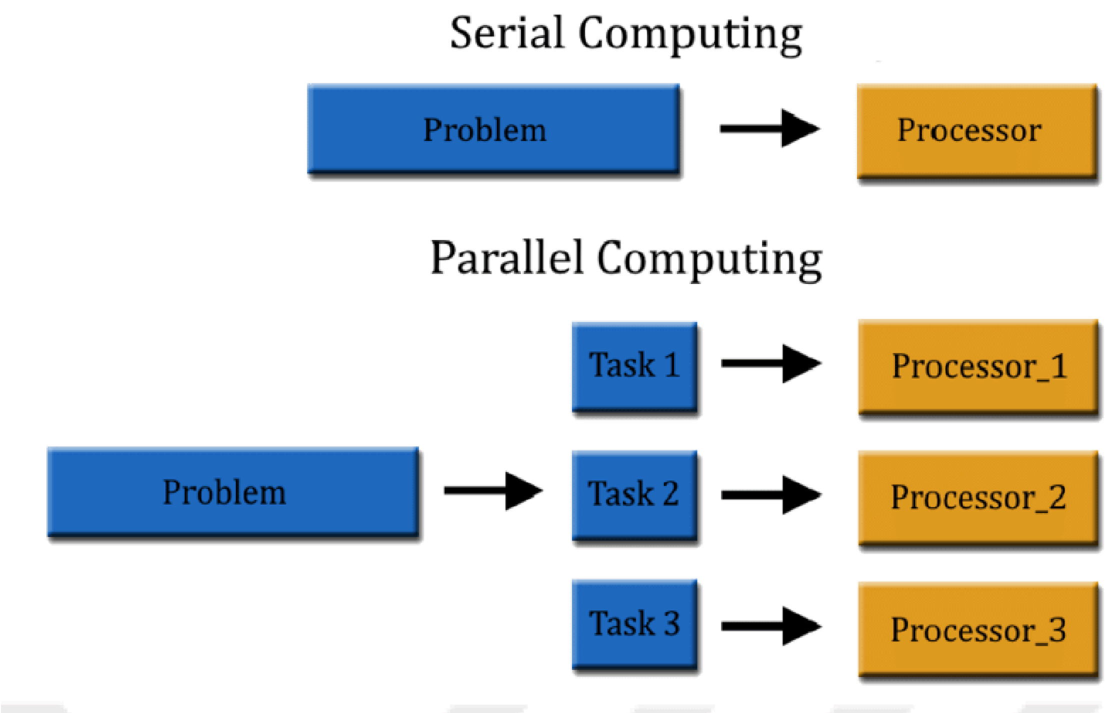


Figure 1. Linear versus parallel data processing (Ortiz, 2020)

Foundation Models are AI systems built upon this transformer architecture. They are trained on enormous datasets and can fulfil a vast range of general tasks. They are utilised as building blocks for facilitating more specialised applications (IBM, 2025b).

2.4.2 Foundation Model Use Cases

While Foundation Models have numerous applications, their most popular use case is Natural Language Processing (NLP). NLP focuses on understanding, interpreting and generating human language, efficiently predicting the next most probable word. This process is done through the employment of text embeddings, the action of turning words into numerical vectors, establishing a mathematical relationship between words.

Furthermore, foundation models using the transformer architecture can be modified and highly specialised to benefit a wide range of fields. Geneformer is a foundational model used for drug target discovery and developmental biology that embeds gene expression data by transforming single-cell RNA sequencing data into dense, high-dimensional vector representations (Theodoris et al., 2023). Traditional transformer architecture can also be modified into vision transformers to accept images as inputs and vectorise its fragments, or patches, the same way NLP does text. Contrastive Language-Image Pre-training (CLIP), developed by OpenAI, combines a vision transformer with text transformers to map images and text into a shared, high-dimensional vector space (OpenAI, 2021). Further, Video Action Transformer models like the one introduced by Girdhar (2019) take an input in the form of multi-modal data, embed the surrounding context, and predict the most probable action or behaviour. Lastly, RT-2, a Vision-Language-Action (VLA) model, combines a visual transformer with text transformers that take visual perceptions and turn them into robotic motor commands (Google DeepMind, 2023). However, this list is not exhaustive, and the use cases of foundational models are constantly expanding across various fields.

2.4.3 Anticip8 as a Behavioural Predictor

A major component of the technology used in this project comes from the use of a service called Anticip8, created by the company Fountech AI. While Fountech is strict on disclosing full details about their training data sources, proprietary datasets, or internal model parameters, they provided the following information:

"Anticip8 is a predictive AI system designed to forecast human behavior and decision-making. The core methodology involves analysing behavioral signals, contextual variables, and historical patterns to generate probabilistic predictions about future actions. The system operates on a multi-component architecture: natural language processing for understanding intent and context, behavioral modeling for pattern recognition, and an action prediction layer that synthesises these inputs into forecasts.

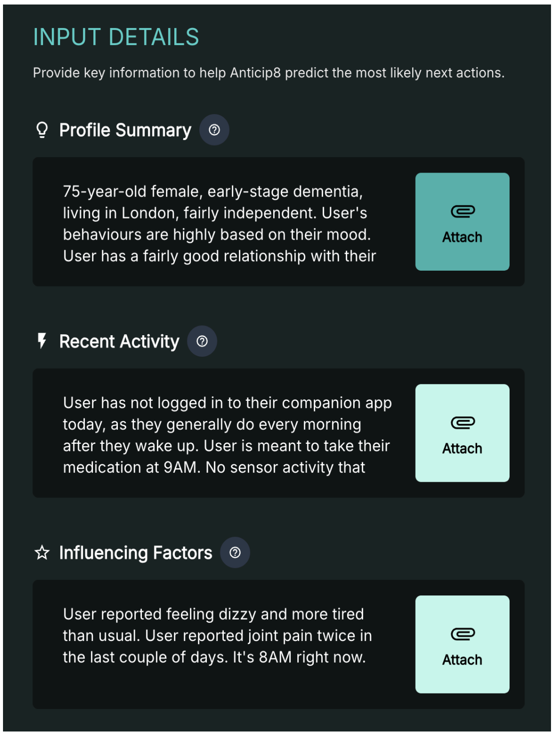
Key technical concepts to reference:

- Behavioral signal processing (extracting meaningful patterns from user interactions)
- Contextual embedding (incorporating situational variables into predictions)
- Temporal modeling (accounting for how behavior evolves over time)
- Probabilistic action forecasting (generating likelihood distributions over possible future actions)

The innovation lies in treating human decision-making as a predictable phenomenon when sufficient behavioral context is available, moving beyond reactive analytics toward genuine anticipation. Unlike traditional recommendation systems that suggest based on similarity, Anticip8 predicts what a specific individual will actually do next based on their unique behavioral fingerprint."

Based on the above information, it can only be speculated how this hybrid architecture transformer works.

While this project's use of Anticip8 will be through their application programming interface (API), they provide a useful playground feature to test their services. The interface allows for the input of all the necessary information to generate and rank the next most probable user actions. The input fields labelled "Profile Summary", "Recent Activity", and "Influencing Factors" *(Figure 2)* aim to provide sufficient context about the user and their surroundings, while the "Next Actions" section *(Figure 3)* lists possible next actions to be ranked (not generated by Anticip8).



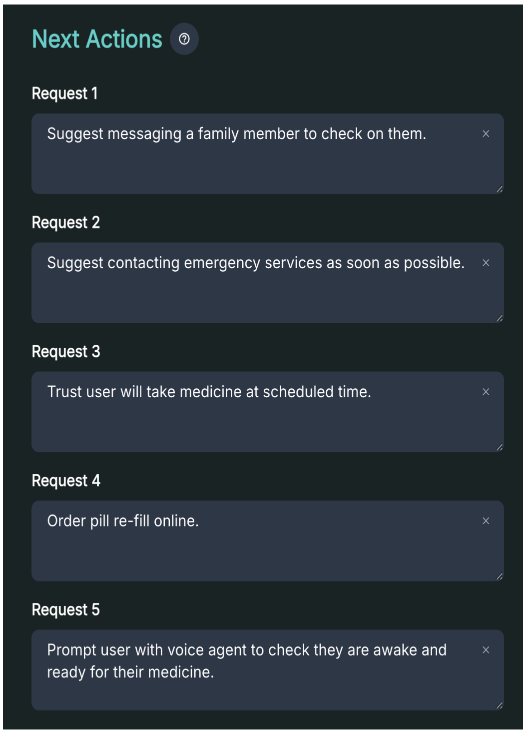


Figure 2. (left) Input details from the playground. Figure 3. (right) Next Actions inputs from the playground.

After Anticip8 processes the provided inputs, it presents an array of both the provided and new actions, ranked from most to least probable *(Figure 4)*.

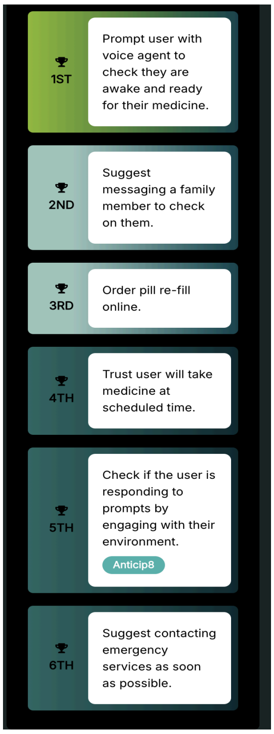


Figure 4. Actions ranked by Anticip8

2.4.4 Agentic Workflows

While Anticip8 provides the needed foresight by predicting a user’s potential actions, risks, or needs (e.g., detecting a high probability of a fall based on uneven pavement), prediction alone is not enough to prolong the independent living of the elderly population. A prediction is only data; the system needs to be able to interpret these predictions and act accordingly. This calls for a technology with the capability to learn and complete tasks in dynamic environments, in order to support the user’s needs when they arise.

An agentic workflow, or an AI “agent”, is a system that uses a large language model to interact with their environment in order to, given a certain degree of autonomy, proactively achieve a goal or complete a given task (Wang et al., 2025). This way, AI agents have the capability to iteratively reason, plan, and execute the provided tools in order to complete an objective. This ability to interpret natural language, choose the correct tool for the task, and execute upon it is the direct opposite of a traditional hard-coded set of steps.

The key concept that drastically separates these two systems is the architectural pattern used, known as the *ReAct pattern*. Often composed of perceptive, reasoning, and acting modules (Yao et al., 2022; Amazon Web Services Documentation, 2026): The perceptive module serves as an ingestion layer, transforming multimodal data into structured representations that inform reasoning. The reasoning module is the cognitive element of this system, integrating memory, knowledge representations, goal structuring, and planning to create informed decisions on how to proceed. Through trade-off evaluation, the decision-making process allows the workflow to stay on goal, choose appropriate actions and close the gap between perception and behaviour. Finally, the acting module executes the workflow’s selected decision by, in our case, calling a predefined function programmatically. After execution, the generated outcome is looped back to the perceptive module, effectively updating the workflow’s knowledge base and influencing future decision-making.

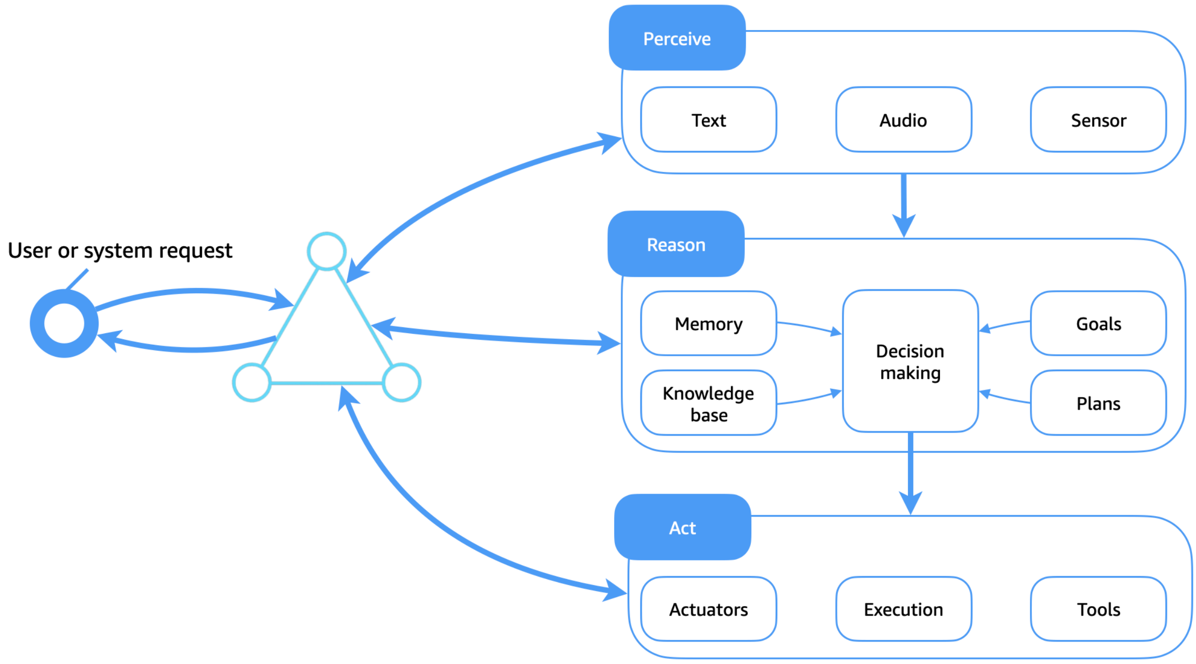


Figure 5. Architectural visualisation of an agentic reasoning pattern (AWS, 2026)

However, it is important to note that agentic AI systems in charge of large tasks perform best when these tasks are broken down, so that multiple agentic workflows can handle different elements of the task (Kim et al., 2025). The aim of doing this is to specialise each workflow and tailor its prompt to the individual task, or type of task, it is meant to perform. When correctly implemented, multi-agent systems tend to outperform single-agent systems and benefit from a lower hallucination rate, more effective self-correction, overall latency performance, and task completion rate performance. This is due to the introduction of coordination mechanisms, structured communication protocols, and the ability to share learned experiences (IBM, 2025c; Kim et al., 2025).

2.5 Conclusion

The day-to-day life of an elderly person is dynamic and requires a system that can keep up with its unpredictable nature. AI agents fit these requirements to act as a bridge between Anticip8's predictions and the user's reality. The main purpose of this project is to test and bring a use case to life of a technology that reshapes the self-attention mechanism that powers transformers to process human habits and actions based on their unique behavioural fingerprint, instead of syntactical information. By combining the predictive capabilities of action transformers and the responsiveness efficiency of agentic workflows, this project attempts to build a system that not only monitors the elderly in their journeys but also proactively and wisely collaborates to prolong their independent life.

3. The Product

This chapter focuses on explaining in detail the technical implementation and product design of SafeStep. SafeStep is a journey planning and assistant application developed to proactively help users navigate the TfL public transport system. Built on a FastAPI Python backend, the system is composed of two main data pipelines. First, a data ingestion pipeline focusing on gathering and processing appointment email data, turning it into structured Google Calendar events. Then, a journey planner pipeline is designed to analyse the user's journey and set up prevention measures to keep the user on track during their trips. Furthermore, the user interface is developed as a React Native application. This application features a dedicated visualisation tool on the web and an interactive map on mobile for users to take with them during their journeys.

3.1 FastAPI Backend

3.1.1 Backend Architecture

The SafeStep backend is designed to support the significantly time-consuming processes that AI applications perform. Given the inherent latency associated with LLM reasoning loops and audio generation employed to fulfil journey planning, a traditional synchronous web server would result in a poor user experience. To mitigate this, the system uses an asynchronous, event-based architecture to separate the user-facing web server from the heavy AI workflows. This makes sure the application stays responsive and allows the user to interact with the system freely.

The entry point for most user interactions is a RESTful API. When the mobile application sends an HTTP request, the payload is passed to the internal routing logic, and an acknowledgement in the form of a 200 OK response is sent back to the client. By handling requests this way, the mobile application's UI is freed, allowing immediate user interaction while the AI processes run in the background.

Internally, information is communicated through Redis, implemented as a message broker with two crucial functions:

- **Task Queues:** When a request requires significant AI processing, such as tool usage or heavy graph calculations, the system formats the request into a standardised packet and pushes it into a task queue.

- **Pub/Sub:** Once a background task is completed, it needs a way to be communicated back. For this, Redis pub/sub channels are created for each user session, allowing the background task to be broadcast when needed.

Since the initial HTTP request is closed immediately, the server requires a separate way to communicate the finished responses back to the user. SafeStep employs persistent WebSockets to fulfil this need. Another way SafeStep takes advantage of WebSocket's bidirectional communication is during specific situations where automatic messages need to be streamlined back to the agentic workflow. The most frequent case in the application is sending the user's location when they are perceived to be off track in order to trigger a route recalculation.

Finally, the system is packed in a Docker image, deployed and hosted in an Amazon Web Services (AWS) EC2 instance. Hosting the backend in AWS provides a robust, scalable cloud environment. This ensures that the WebSocket connections and asynchronous task queues remain stable, secure, and continuously accessible to the client in real-time, regardless of the user's physical location.

3.1.2 Agentic Workflows

When planning the general SafeStep workflows, the implementation of a multi-agent system in combination with Anticip8 seemed to be the correct choice. Initially, the system would have a central orchestrator agent acting as the primary gateway for all user interactions, task delegation, and agent supervision. However, after the first implementation and further reflection, this led to a significant bottleneck in performance. Having every user interaction, from a tool usage to a simple interaction, go through several LLM calls meant a considerable delay.

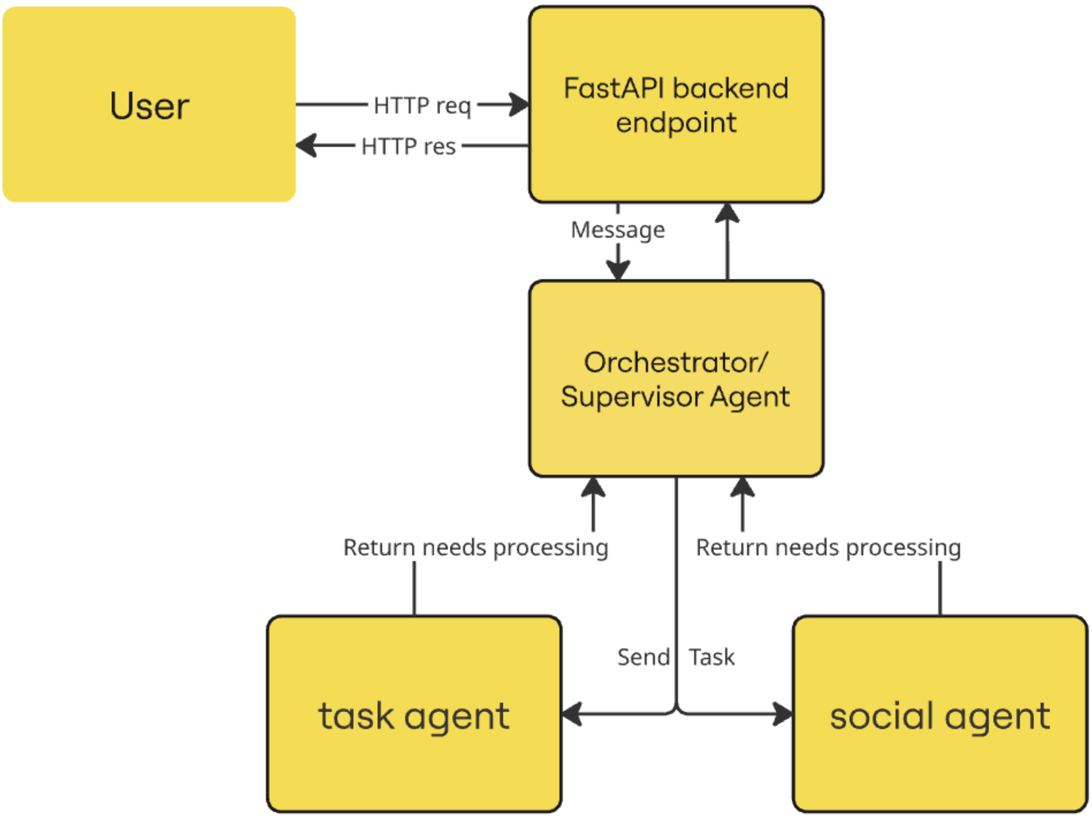


Figure 6 initial implementation of backend architecture

Instead, the rework of the system shifted towards the following strategy. When dealing with user interactions, the backend routes their messages into our agentic workflow. To avoid performance bottlenecks and ensure workflows have well-defined responsibilities, the agentic system is composed of two main workflow components, a user/front-facing agent and an orchestrator/thinking agent. To ensure standardised communication between these agents, the system uses the FIPA-ACL (Agent Communication Language) protocol, as it is widely used across systems.

The front-facing agent acts as a triage agent, receiving all messages and classifying them based on their intent. If the message is a simple interaction with the system that does not require a tool usage, this agent would handle said interaction and return a response with a soft, caring tone. Otherwise, if the message was intended for the system to perform an action through tool usage, the front-facing agent would dispatch the request to our orchestrator agent with tool access. This autonomous decision-making allows the system to perform actions based on task importance, tailored to the specific user request.

When the message requires a complex action to perform, the client-facing agent composes a packet object containing the task information and pushes it into the orchestrator queue using Redis. Subsequently, it returns a response to the front end so the user knows the message has been processed.

The orchestrator agent is initiated by an execution file that listens to the Redis queue assigned to the agent. When the front-facing agent's packet is received, an instance of the orchestrator agent is created, and the message is delivered. This is the workflow with most agency, employing a ReAct pattern that performs an iterative loop of thought, action and further observation to plan and execute functions to fulfil a given request. This agent has access to a diverse toolset, including Google Maps for calculating navigation routes, email sending to reschedule appointments, and a user interaction tool.

This workflow uses the OpenAI Completions API, which requires manual message injection to reliably provide context. To do so, the system employs a ConversationManager, a file that stores relevant messages and the agent's own reasoning, into a database table for constant access. Processes are labelled with an ID corresponding to the current task.

An important aspect when developing workflows with increased amounts of agency is keeping the user in the loop when dealing with important task delegation. To fulfil this aspect, the system uses the user interaction tool, allowing users to contribute to the decision-making process. When a process requires it, the system sends a message directly to the user and terminates its current run, only resuming when a new message with the corresponding ID arrives. After any given process is finished and ready to be sent, the system turns its response into an audio version, uploads the .mp3 file into a database bucket, and prepares a package intended for the user. This package is returned to the execution file and subsequently sent to the user interface.

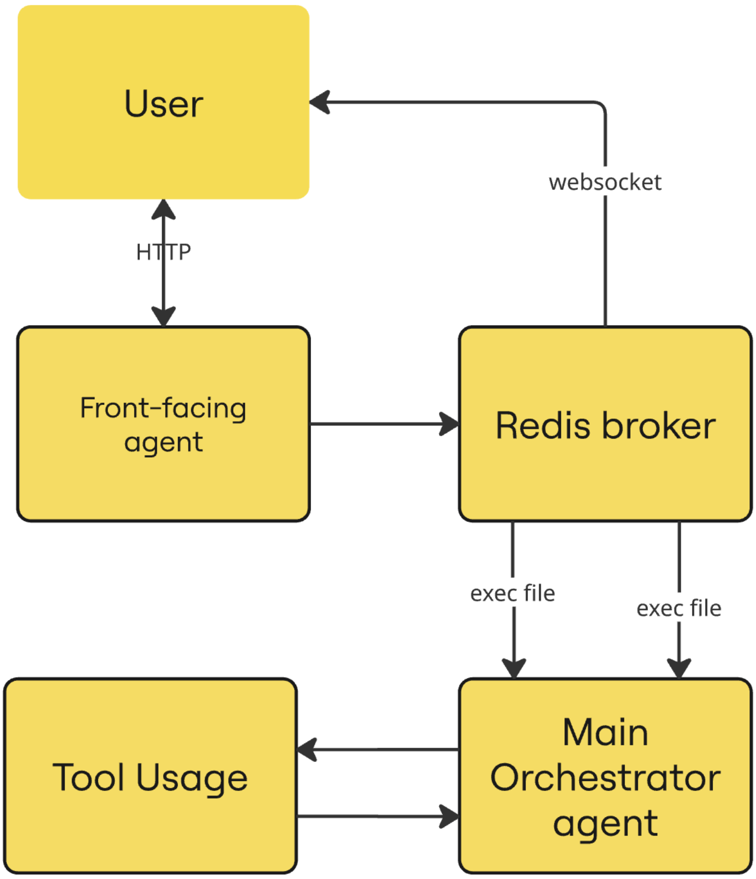


Figure 7 illustrates how user messages travel across the system.

3.1.3 Data Pipelines

For SafeStep to fully perform as intended, robust data ingestion, manipulation, and processing steps are necessary. The system relies heavily on two data pipelines working closely together. The first handles administrative data ingestion, organising unstructured email information into managed appointment events. The second aids in route generation, proactive planning, and incident prevention. Together, these processes eliminate the cognitive overhead required to schedule, organise, and plan appointment-related affairs, seamlessly connecting the user's digital schedule with their real-world journeys.

3.1.3.1 Email Ingestion and Calendar Event Manager Pipeline

This is an event-driven ingestion pipeline that fetches user unstructured email data and turns it into appointment-related calendar events. The purpose of this pipeline is to establish a single source of truth for the user's medical appointment journeys. While the Journey Planner pipeline can be seen as the most relevant for the application itself, this pipeline enables the system's reactive data visibility mechanism. Without this pipeline, users would have to rely on direct communication through voice or text. This pipeline is composed of two main components, an Email Processor, in charge of fetching and chunking emails, paired with a Calendar Event Manager, who deals with booking, rescheduling and cancelling calendar events.

Additionally, this pipeline sets the ground for future work on the application by generating text embeddings and storing them in a database table. While not currently implemented, this could give the system the capability to proactively provide appointment information or answer any questions regarding an appointment the user could have.

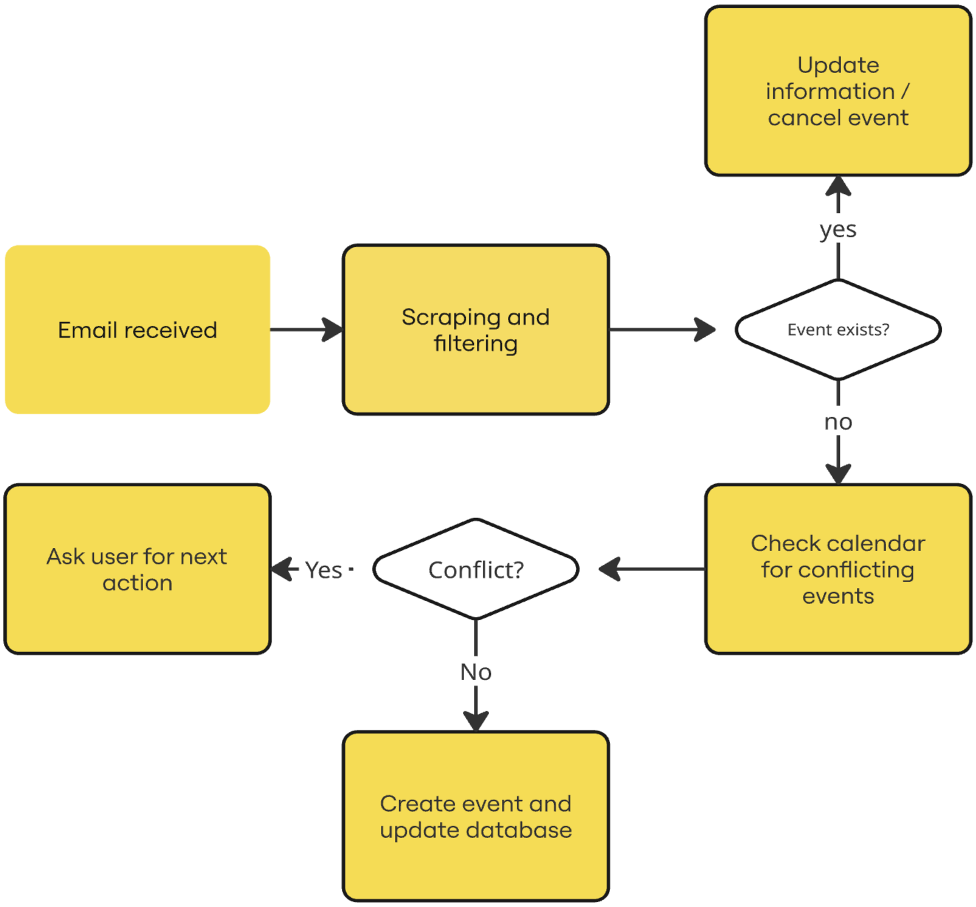


Figure 8 Gmail and Google Calendar pipeline data flow

As shown in the graph above, the steps the pipeline puts data through are as follows:

1. It fetches the last 10 emails in order to make sure no emails are missed. Subsequently, appointment-related unread emails are filtered and processed by chunking and generating text embeddings.
2. New emails are further processed to extract key information such as location, date and time, sender, and intent. This information is stored in an email database.
3. Once the emails are processed, the pipeline extracts useful information and analyses email intent.
   - If the email is identified as regarding a new appointment, it checks if there are any events in the calendar during the start and end times of the current event being processed. If the user is free, the system creates a new event object. Otherwise, an issue is placed in the orchestrator

agent’s queue so it drafts a rescheduling email, shows the user and asks for their approval. If the email is sent, the process is repeated once a response email with the same thread ID is received.

- If the email is intended to reschedule an existing appointment, the system makes the necessary changes and updates the calendar. Future refactors of this feature will include checking user availability for the new date and time, treating the new event as a new appointment and sending a rescheduling email in case of the user’s calendar being compromised.
- Finally, if the email is intended to cancel an appointment, the system checks the calendar and promptly cancels the corresponding appointment.

3.1.3.2 Journey Planner Pipeline

This pipeline is at the core of the application functionality and is run on virtually every interaction with the application. It is in charge of not just doing a route calculation, but based on the user’s profile and their specific vulnerabilities, foresees failure situations and promptly generates preventions for them. To do so, the data pipeline receives required parameters such as origin, destination, the user's profile and, optionally, LLM specifics and returns a directed graph holding an array of steps, each with its potential failure points and preventions with varying degrees of efficiency.

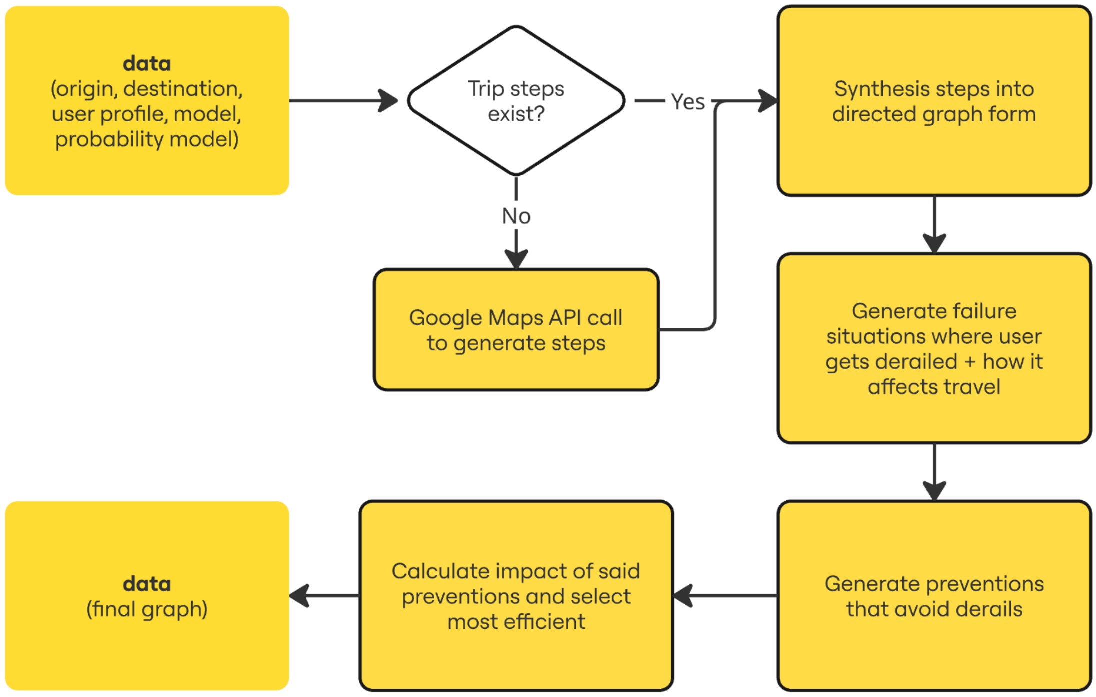


Figure 9 Journey Planner pipeline data flow

As shown in the graph above, the steps the pipeline puts data through are as follows:

1. Journey steps are calculated using the origin and destination, and subsequently transformed into a structured directed graph.
2. For each of the directed graph nodes, up to four new failure nodes are generated. Alongside this failure, information is attached to their generated probability score, representing the likelihood of the user deviating from the "correct" path into this specific failure.
3. After that, tailored preventions are generated, alongside text and audio descriptions to ensure a more accessible delivery.
4. With the complete graph generated, a new LLM call is made, intended to recalculate each probability score. This step measures how much each prevention reduces the likelihood of the user making an incorrect decision.
5. Finally, the prevention that yielded the highest likelihood of the user going in the intended direction is highlighted, and the final graph structure is returned to be displayed in either the web or native versions of the map.

3.1.4 Database

SafeStep uses Supabase as its database and Auth service provider. Incorporating both features allows systems to take full advantage of their services, given that they are both built into the same PostgreSQL instance. This allows both services to communicate natively, simplifying architecture and enhancing security without the need for extra boilerplate code.

A significant benefit is required Row Level Security (RLS) polices, rules that allow the database to filter results to only return information compliant with the written policy. RLS policies can be used to allow users to access and edit their own chats, submit bug reports, and allow the system itself to access, add, and edit email information.

```
create policy "Users can submit bugs"
on public.bug_report
for insert
with check (auth.uid() = user_id);
```

Figure 10 An example of RLS policy declaration

Additionally, the system leverages PostgreSQL's functions and triggers to automate significant processes such as assigning a random base profile to each new user on creation.

To conclude the database section, find a comprehensive view and explanation of the most frequently used tables throughout the application.

- **User_credentials:** This table holds user Google credentials needed to manage sessions, such as access tokens for initial logins and refresh tokens for subsequent automatic logins. Additionally, this table holds the last updated and token expiry information to facilitate the creation of a new Google Credentials object using the refresh token.

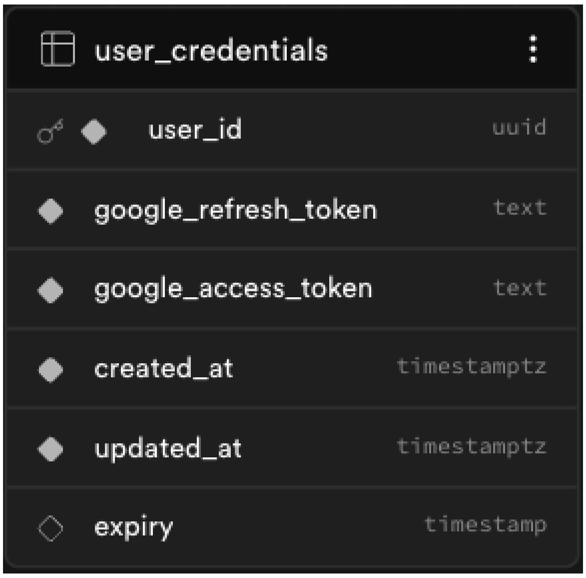


Figure 11 The user_credentials table

- **User_derail_coords:** This table holds information about *where* and *when* the user was derailed from their routes. This not only includes their coordinates, but also their destination, timestamp, and incident ID.

Figure 12 The user_derails_coords table

- **Emails:** This table holds all information about user appointment emails. It is crucial for appointment tracking, used to send rescheduling emails, and fetch information required to orchestrate future journeys.

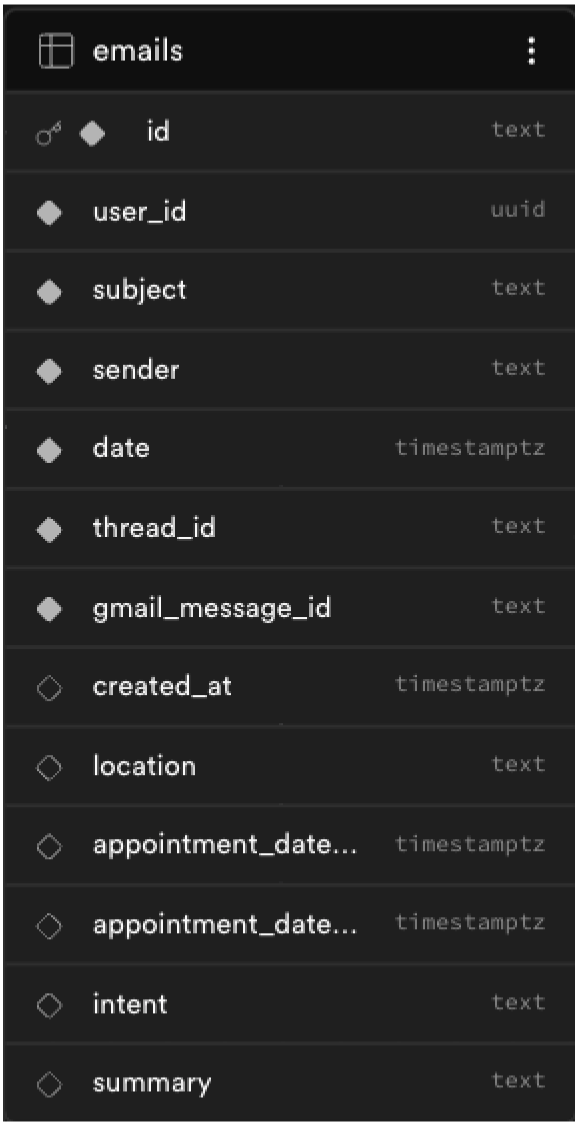


Figure 13 The emails table

- **Chats:** This table holds all user messages, where the messages column is composed of JSON objects with individual user and system message information.

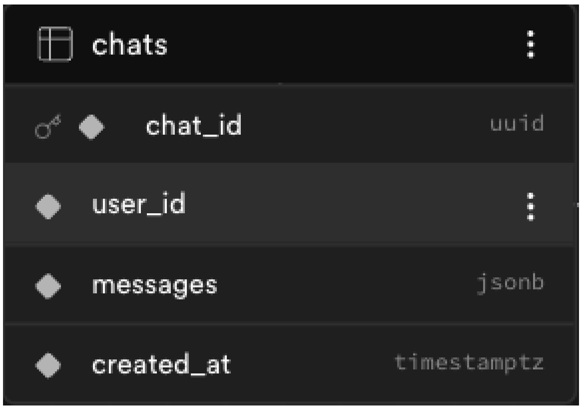


Figure 14 The chats table

- **Users:** This table holds user information, composed of email, user ID, and profile information. Profile information is inherited from the base_profiles table at random when a user is created, composed of separate JSON objects required to perform Anticip8 API calls. Additionally, separating these features eases the process of updating travel information from the instrumentation insights.

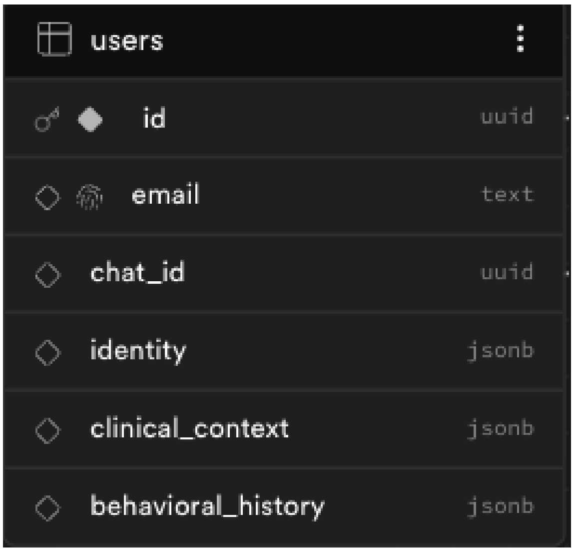


Figure 15 The users table

- **Base_profiles:** This table holds the two base profiles that new users inherit, focusing on either dementia or frailty. The columns are composed of JSON objects for easy access. Identity contains personal information, such as name, age, and living situation. Clinical_context includes condition-specific information. Behavioral_history contains relevant and profile-specific travel information. Both Clinical_context and Behavioral_history are used to inform and aid AI models in making the best possible predictions.

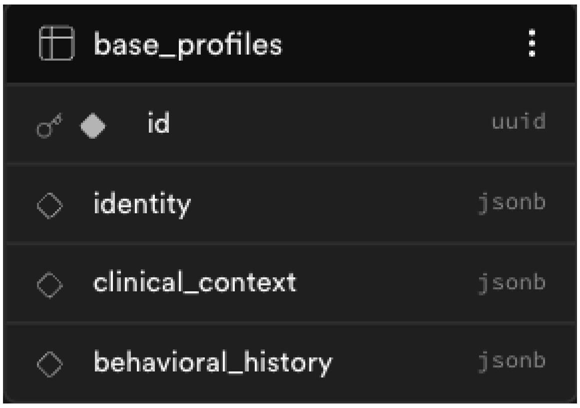


Figure 16 The base_profiles table

- **Trip_history:** This table holds information useful to analyse and visualise the testing phase data. It holds useful information that describes journeys such as user, data and time, origin and destination, trip success information, trip recalculation information and the number of prevention audio plays (used for instrumentation).

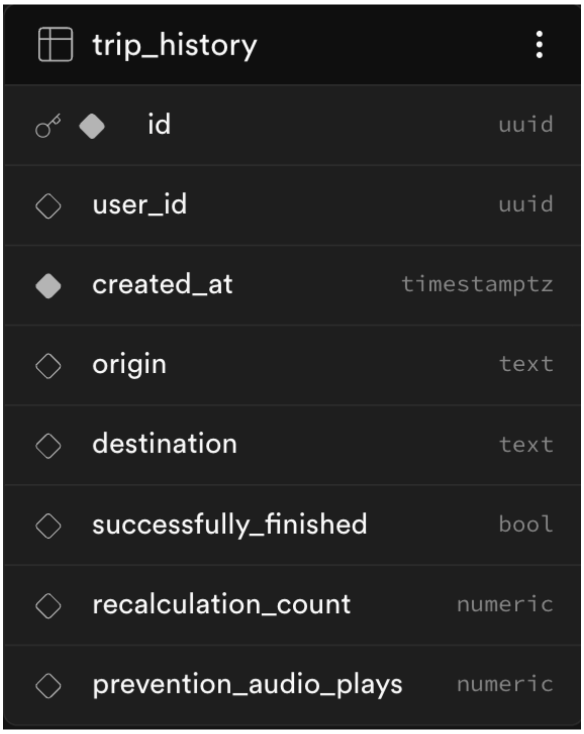


Figure 17 The trip_history table

- **Bug_report:** This table holds bug report information, composed of user_id, created_at timestamp and an explanation of the bug in plain text form.

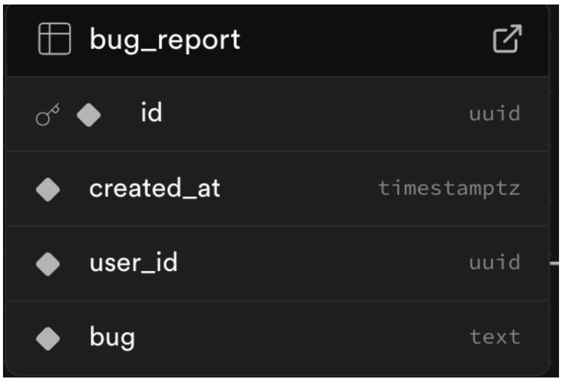


Figure 18 The bug_report table

3.2 React Native Application

React native is a JavaScript framework developed by Meta in 2015 that follows the same modular principles as ReactJS. It is widely used by mobile developers and engineers since most of the code that is written in React Native can be shared between platforms, which simplifies simultaneous development for iOS, Android and web. This is the main reason this project has been built using React native, despite the development being focused on iOS, future work includes porting and optimising the Android environment.

3.2.1 Pages

SafeStep users interact with the application through three main "pages" or "views".

- **Index.tsx:** This file is the first page the user sees and currently hosts the login logic and, once logged in, a dashboard with insights for the user and their caregiver. While not yet implemented, future works on this file include the implementation of a full-featured dashboard displaying users' calendar information related to medical appointments, including system intersections such as rescheduling and cancellations and other useful features for both users and caregivers, such as reminders and device vitals.

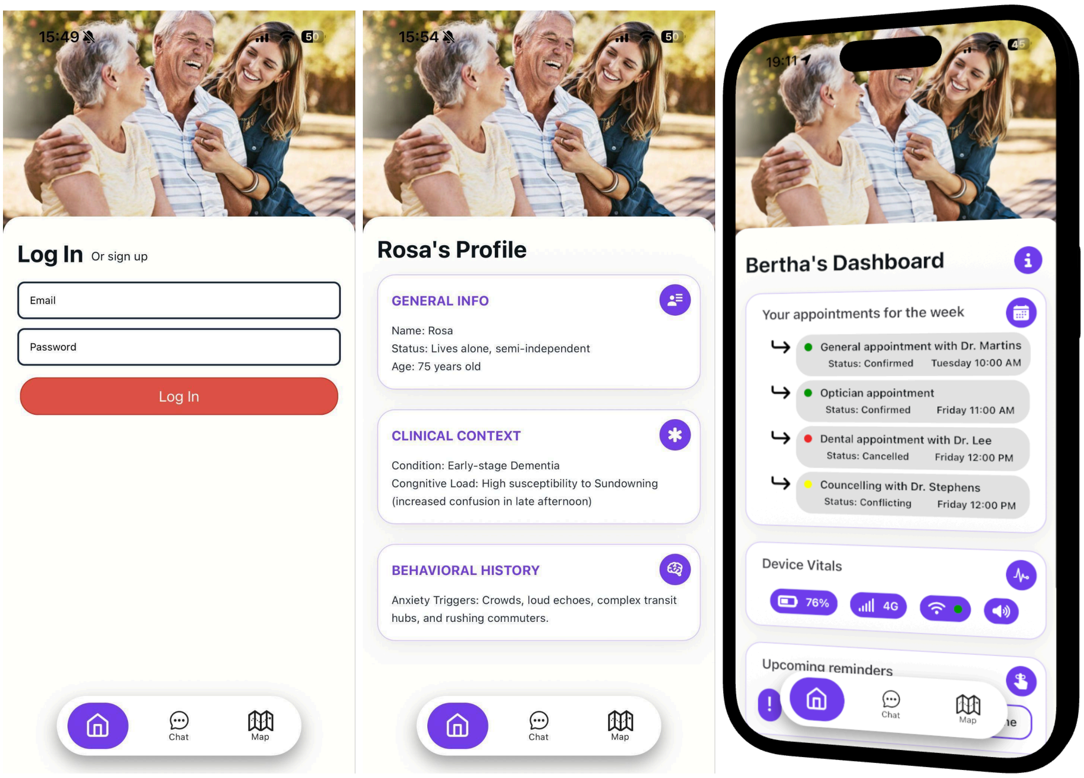


Figure 19 (left) log in screen, Figure 20 (centre) Dashboard placeholder showing profile, Figure 21 (right) showing future works on dashboard example.

- **Chat.tsx:** This file allows the user to communicate directly with the system, whether that's for initiating the journey planning feature before a trip, green-lighting an appointment rescheduling email, or a simple social interaction with the LLM. When communicating, the user has the option to do so through a written or voice message. Subsequently, responses are both available in a traditional chat fashion and immediately played out loud with a calm, caring tone. This is achieved through speech-to-text of the sent audio file and text-to-speech audio file generation in the backend.

Figure 22 (left), Figure 23 (centre) and Figure 24 (right) Chat states

- **Map.tsx:** This file hosts the logic for the two core journey functionalities. On the web, it displays the journey visualisation tool, and on mobile, it displays a native map showing current journey-related information. Map display is implemented using the "react-native-maps" library. Additionally, this file allows the user to report bugs encountered during their trips. All interfaces, their functionality, and screenshots will be explained in depth in the 3.3 Journey Planning section.

3.2.1 Main Components

The modular nature of the React framework allows for development with SOLID principles in mind. This means that while pages are the files hosting the code for the user's interaction, they are composed of individual, reusable UI components mostly in charge of a single responsibility. This section is dedicated to briefly highlighting some of the most relevant components across the three main pages.

- **VoiceComponent.tsx:** This component renders the voice chat interface that, paired with the written chat interface, allows the user to communicate with the system.

- **ContentModal.tsx:** This versatile component acts as what is called a base component. Base components are components that allow for higher flexibility by setting up basic design features that can be used throughout an application. Customisation comes in the form of injecting content directly into the form of children. This is a base modal component, used to model both the arrival modal and bug reporting modal.

- **MapComponent.web.tsx/MapComponent.native.tsx:** These two components represent the mobile and web variations of how the app displays data returned by the backend when requesting a journey calculation. The reason these are separated into distinct components is because of how drastically different they work. On the web, this translates to the visualisation tool, displaying a graph of pure nodes and edges automatically laid out in the UI. In the mobile version, this is a map displaying the user's location and the route's polylines. These implementations and their uses are discussed further in the upcoming sections.

- **CustomNode.tsx:** This component encapsulates the logic behind creating and rendering nodes and edges formatted by the mapDataTransformer.ts file. It leverages the graph rendering library “@xyflow/react”.

- **JourneyInformationComponent.tsx:** This component encapsulates journey polyline logic and is in charge of displaying the journey information section. The journey information section is composed of an overview of the user's location in relation to the general trip and text and audio prevention display. This component de-clutters the native map component, improving both files' readability.

- **ProfileDashboard.tsx:** This component displays the user's dashboard. While it currently only displays key user profile information such as age, condition and anxiety concerns, in future iterations it will display more useful information about the user's appointments and trip statistics, useful for both the user and caregiver.

3.2.2 Hooks

React hooks are pieces of reusable logic that take advantage of React features and give developers access to React's internal memory. The most commonly used React hooks are useState to set and manage state, useEffect to watch variables and manage side effects, and useMemo to memoize values, among others. This section will not focus on these, rather, it will briefly highlight custom hooks developed to facilitate the implementation and distribution of the application's logic.

- **useRouteMonitor.ts:** This hook is the most significant hook in the application. It handles all side effects related to user location other than the location tracking itself. Functionalities include:
    - Keeping track of user derailment.
    - Keeping track of the current step and the subsequent polyline the user is on
    - Syncing the summarised steps from the "graph" containing the best prevention text and audio file link fetched from the back-end with the google maps api steps for the user to follow.
    - Decode and set the individual and general polylines' state.
    - Reset states when the user reaches their destination.

- **useLocationTracker.ts:** This hook is in charge of requesting user location permissions, keeping location state and tracking location with high accuracy every two seconds. In order to track location, this hook leverages the "expo-location" library.

- **useAuthStore.ts:** This hook is how the application accesses global state, implemented through the global state management library "Zustand". Global state allows every file in the application to access state setters and getters. Initially used to manage user auth state, it was later used to hold individual and general polylines' state. Alternatively, other common ways of facilitating state are prop drilling, the React context API, or other libraries such as Redux.

- **useGraphLayout.ts:** This hook handles the automatic node/edge layout used by the visualisation tool. To do so, it leverages the layout library "d3-hirearchy".

- **WebSocketManager.ts:** While not a hook per se, this file leverages the contextAPI distributing WebSocket logic and granting access to the connection to both Chat.tsx and Map.tsx. It establishes and manages the WebSocket connection and re-connection, allowing for bidirectional communication over a single, persistent connection between client and server.

3.2.4 Utils and Helper Functions

In order to keep components as lean as possible, only containing necessary logic and rendering, there was a growing need to move general or miscellaneous logic into its own file, extracted only when needed. This section is intended to highlight the most relevant files containing logic used by main components that do not leverage React's functionality and, therefore, are not encapsulated into a Hook.

- **locationUtils.ts:** This file handles the logic used to track the user's position in relation to the polyline, whether that's the distance between two points or the distance between the user's location and a general segment. This file heavily relies on functions from this website.

- **mapDataTransformer.ts:** This file contains logic related to turning the directed graph data returned by the backend into nodes and edges for the web visualisation tool to display.

3.2.3 Bug Reporting and Fixing

As previously mentioned, during their journeys, users are given the opportunity to flag and report bugs. These bugs are stored directly in a database table that was checked daily during the testing period. The idea of having a bug report feature is rooted in implementing an iterative development paradigm that leverages user testing experience to pinpoint edge cases where improvement is needed.

Users can find the bug report button placed in the top right area of the map page. When tapped, a modal with a large text input box is displayed alongside a submission button. Bug descriptions are stored in a database alongside relevant information such as user ID and a created_at timestamp.

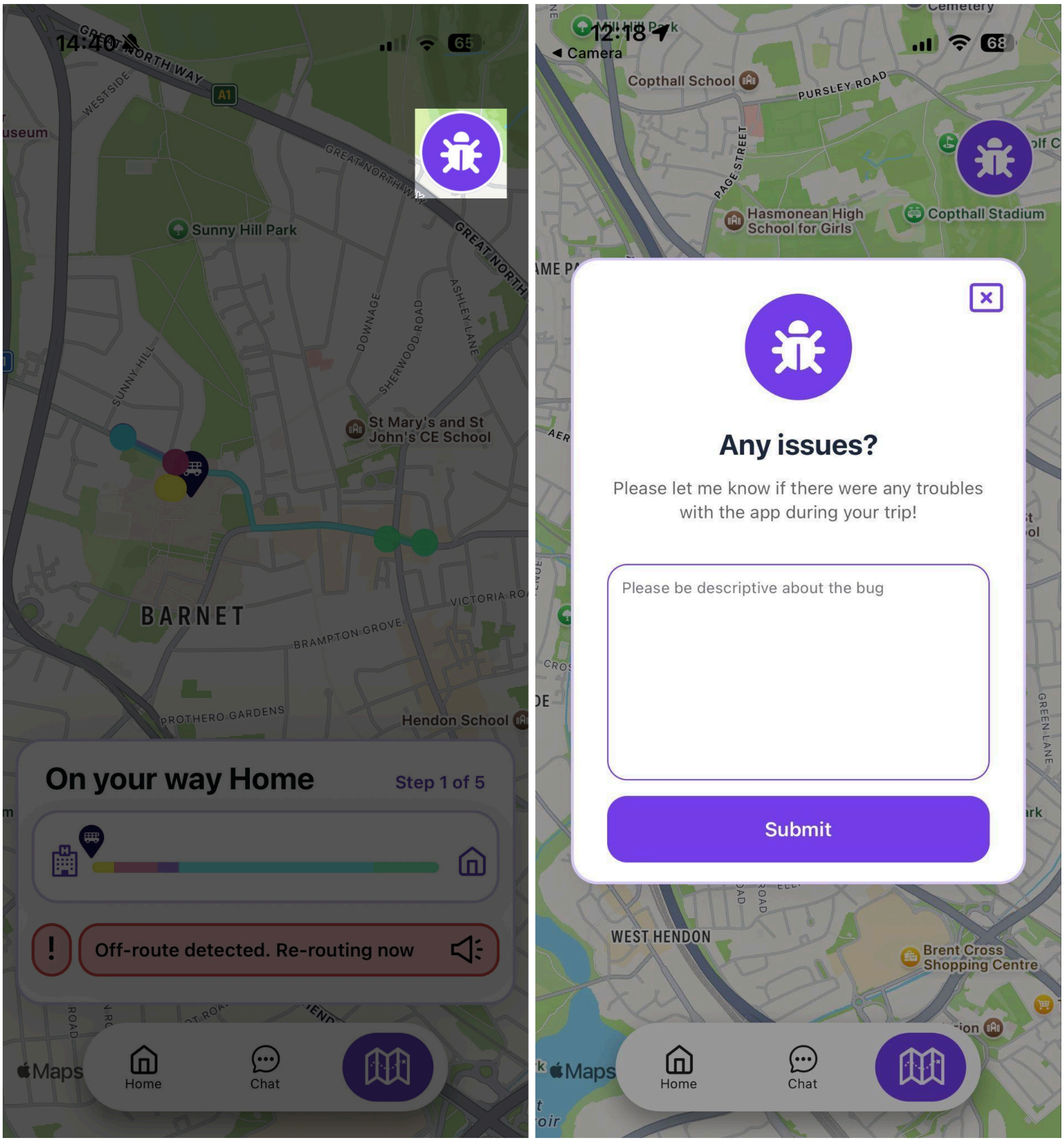


Figure 25 (left), figure 26 (right) showing bug reporting feature

An example of this feature being used and a subsequent bug fix is:

"Sometimes when I take a bus, and I'm about to get to my destination, I get the sign telling me I arrived even before getting to my last stop."

After careful consideration and attempts to replicate the bug, it was concluded that the issue was rooted in the location tracking mechanism. When cross-checking the user's location and their final destination, the system would flag the user's journey as successful if these two coordinates were within a five meter radius. This disregards cases where a user's bus stop goes past their destination and requires the user to walk back in order to arrive. To solve this issue, the system was modified so that, in order to flag the trip as finished, the user should have previously reached the penultimate step's last coordinates.

After implementing and testing this fix, no user has reported this bug during the testing period.

Figure 27 (left) shows the dashboard. Figure 28 (centre), featuring voice chat. Figure 29. (right) showing map features.

3.3 Journey Planner

To achieve the system’s main task of calculating the user’s journey through London's public transport system, leveraging Anticip8’s predicting capabilities, it employs a linear flow pipeline paired with a visualisation tool and further integration with the mobile application.

The core pipeline is designed to generate and transform transit data in the shape of journey steps into a personalised journey graph. The process follows three structured phases. The first phase focuses on generating the journey's travel steps based on its origin and destination, and its further conversion into directed graph form, with each step clearly represented by a node. The second phase leverages the user’s profile and journey nodes to generate distinct failure points that specifically match the user’s limitations. Additionally, this phase generates a numerical representation of how likely it is for the user to follow the correct path or failure nodes, followed by a set of interventions designed to keep the user on the correct path. The last phase is intended to evaluate how effective these interventions are, dynamically adjusting the probability weights of each correct and failure node to reflect the intervention’s effect. The pipeline's output is a structured graph where each node object includes an array of risks and their subsequent interventions.

3.3.1 Visualisation Tool

To evaluate the pipeline's efficiency in addition to aiding in journey planning and supervision, a web-based visualisation tool was developed. This feature allows users to select the trip’s origin, destination and the models tasked with the pipeline’s generative steps. The selectable models have unique but equally important roles in the process. The first model performs the workload described in phase two, i.e., it receives the user’s profile and an array of sequential journey steps that represent the correct path the user must take to arrive at their desired destination. Given this information, for each travel step, it generates possible failure points tailored to the user’s individual characteristics and limitations. Following the identification of these failure points, the model weights the probability of the user advancing through each possible path and assigns it to each respective node. Subsequently, the system allocates each correct path node a set of interventions meant to reduce the likelihood of deviation and reinforce the user's adherence to the intended path. Once this procedure is finished, the second selected model performs the tasks described in phase three, i.e., it recalculates the probability of the user's journey given the application of each of the generated interventions, adjusting the likelihood of both the correct path and failure nodes.

This testing setup allows for comparative evaluation of how different models and their combinations impact the system's risk estimation and how valuable the intervention strategies are in a personalised context. Furthermore, this tool allows the user to have a holistic perspective of their journey, knowing the planned interventions and their estimated effect on each step of the journey.

After running the feature, this is what the UI shows:

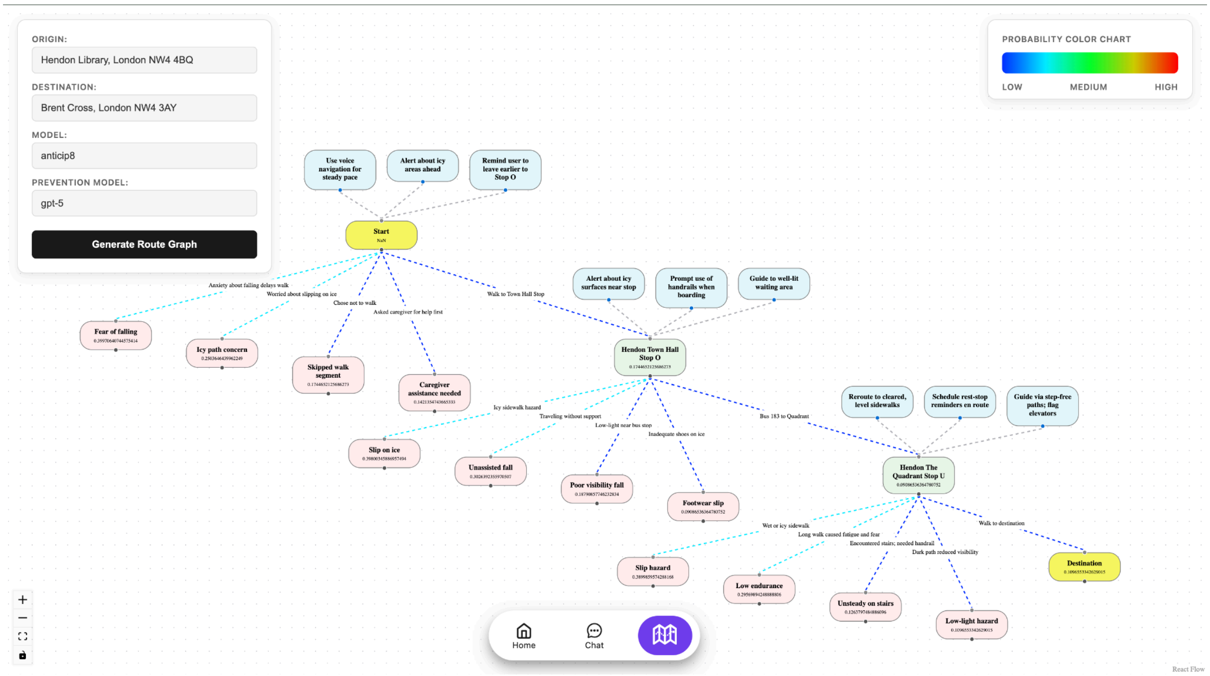


Figure 30 An example of the feature's general result

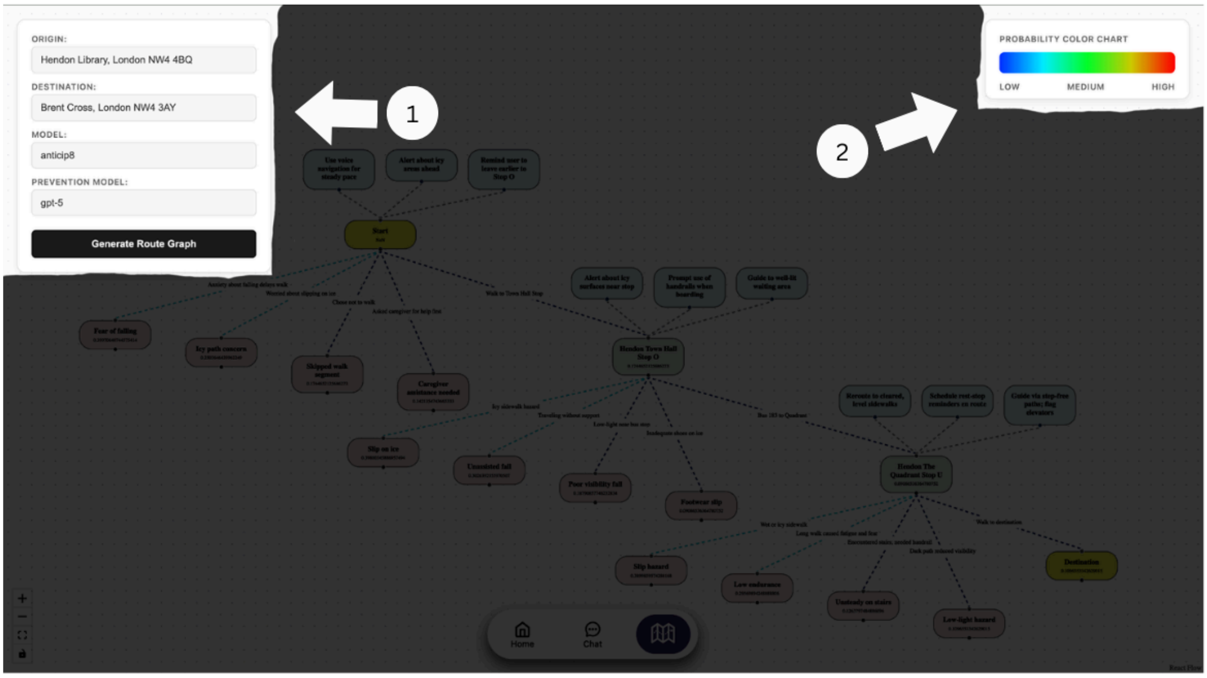


Figure 31 Important UI elements. (1) The input box, with Origin, Destination, Model and Prevention Model input fields. (2) A colour chart that is designed to orient the user when looking at the node edges, in the blue, cyan, green, yellow and red range.

Each journey has a Start and Destination node and inner journey step nodes displayed as follows:

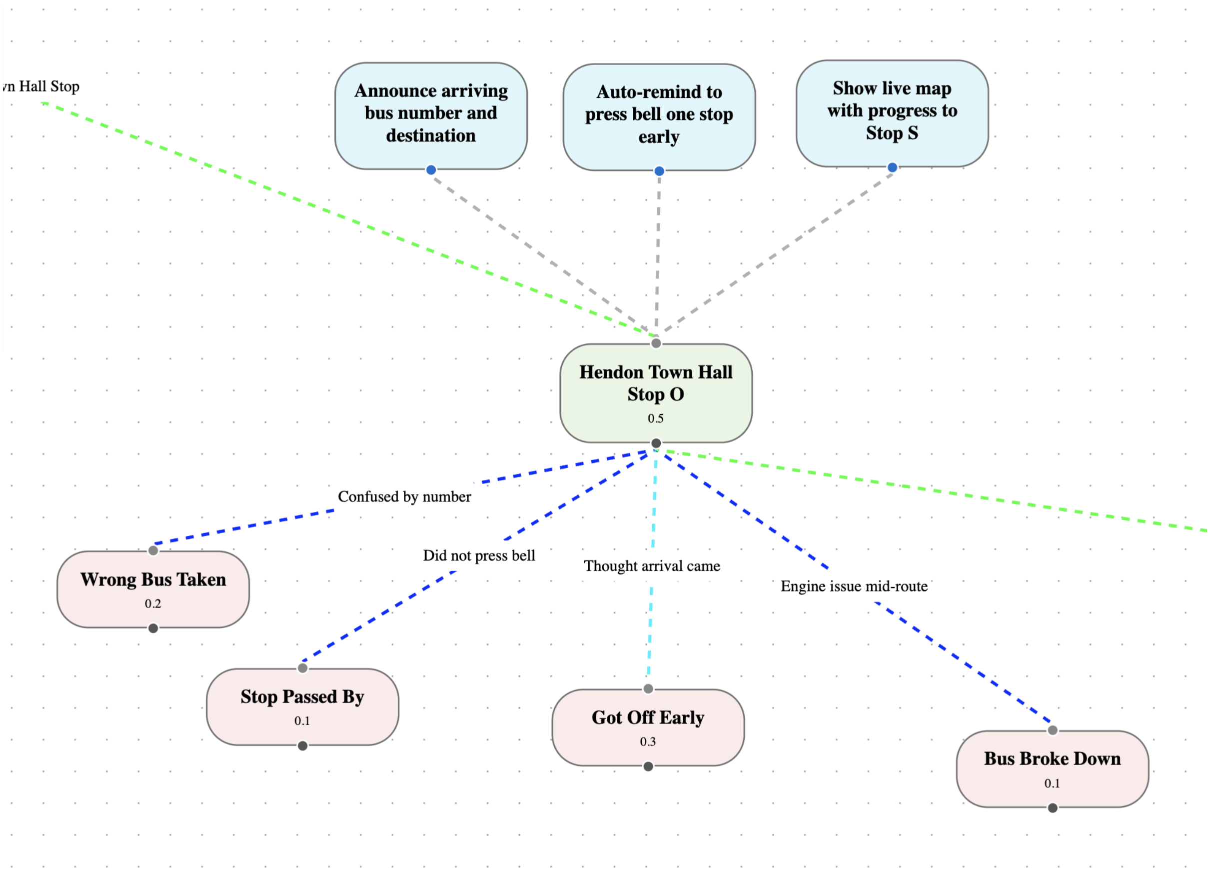


Figure 32 An example of a journey node, interventions and failure nodes

Each journey node will have its own probability, representing the chances of the user transitioning from a previous node to the current node, three prevention nodes connected from the top, and up to four failure nodes stemming from the bottom. These failure nodes are composed of edges, attempting to explain *how* the user reached the failure node, text describing the core issue, and their own probability score.

The edge colours are directly related to the node's probability score, colour-coded following the colour guide. When clicking on a prevention, the application redraws the affected nodes to display their new probabilities.

3.3.2 App Integration

To reflect journey planning and travel steps in the mobile application, the "map" tab displays an interactive map where, given a journey is in place, the user can see a breakdown of their trip. Each step is divided into its own separate colour-coded polyline segments, and an interactive information section guides them through their journey.

To make these functionalities possible, after a travel request, the application receives travel information in the form of the following JSON object:


```
{
  “Content”: {
    "audio_url":
"https://bgkwdvmsnidmyigsrtak.supabase.co/storage/v1/object/sign/audio_files/40422fe1-def2-48c6-a48b-8a6795b27f8f.mp3?token=eyJraWQiOiJzdG9yYWdlLXVybC1zaWduaW5nLWtleV82OWM2NTM4ZC00NjQ0LTQ5ODQtOGY3Mi1lODY2ODliMGNmMjYiLCJhbGciOiJIUzI1NiJ9.eyJ1cmwiOiJhdWRpb19maWxlcy80MDQyMmZlMS1kZWYyLTQ4YzYtYTQ4Yi04YTY3OTViMjdmOGYubXAzIiwiaWF0IjoxNzczNzYzNzg5LCJleHAiOjE3NzM3Njk3ODl9.p4MD_ofeiXV0WXGAiXqkMFsKuwdRwaCjhdXnBDX9I8Y",
   "message": "The trip should take a total of 8 mins and cost around N/A.
    Here are the steps:
    Step 1: Head southeast. It should take 1 min (1 m)
    Step 2: Walk. It should take 1 min (9 m)
    Step 3: Walk. It should take 1 min (7 m)
    Step 4: Walk. It should take 1 min (9 m)
    Step 5: Head east. It should take 2 mins (0.2 km)
    Step 6: Turn left onto Church Rd/A504. It should take 5 mins (0.4 km)"
},
 "graph": {"steps": [[Object], [Object]]},
 "individualPolylines": ["ug{yH`{k@", "ug{yH`{k@HU", "kg{yHjzk@@W",
"ig{yHryk@BU", "eg{yH|xk@V{EDYPO|@cB",
"wc{yHrmk@Ms@KkB?cAC]?kCKaGT@ByBFu@f@oH"],
  "message_id": "7a7f3981-9311-4835-8032-3b5db621cb65",
  "performative": "INFORM",
  "polyline":
"ug{yH`{k@HU@WBUV{EDYPO|@cBMs@KkB?cAC]?kCKaGT@ByBFu@f@oH",
  "receiver": "USER",
  "sender": "OrchestratorAgent",
  "task_id": "4acb5533-8b9d-47f6-a3b8-dccea58a88dc",
  "user_id": "62cc4d43-9576-4bec-a119-418af7893e7c",
  "chat_id": "f6de4e3b-92b9-4bef-8a03-2e4570982ab7"
}
```


The object contains general information such as IDs, receiver and sender, in addition to more relevant information. The graph property contains the processed journey including all preventions for each step, their respective probability calculations, and useful text and audio representations of the prevention that led to the highest success probability. The message property contains the text displayed in the chat interface, explaining all steps, their individual times, and the total cost. The audio_url property contains the audio file URL version of the original message to be played out loud on message arrival. Finally, the polyline and individual_polylines properties contain the general journey polyline and an array of each step's polyline to be displayed and calculate the user's location in relation to their current step and general trip.

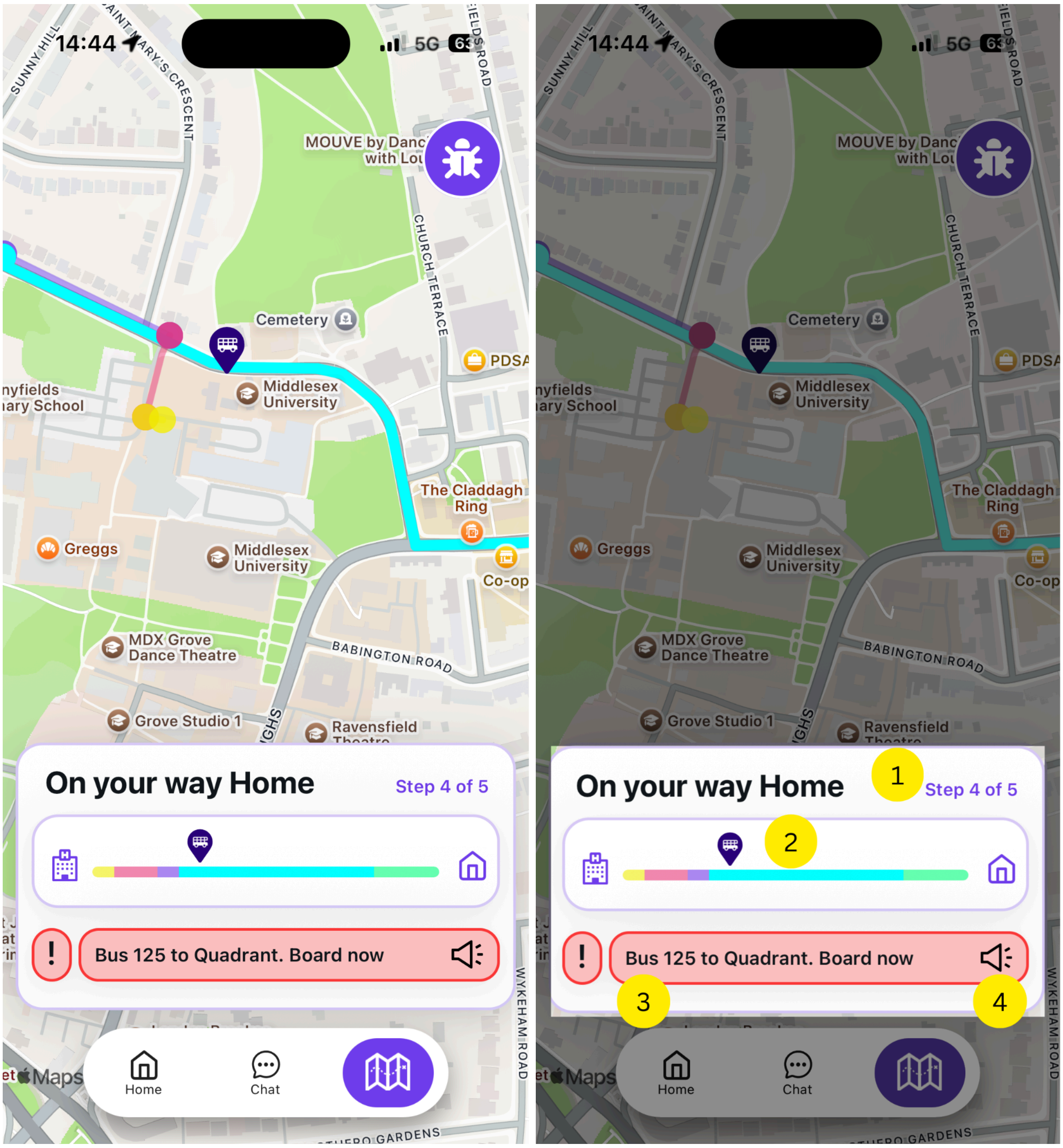


Figure 33 (left) and Figure 34 (right) The map user interface

Figures 33 and 34 above show the map user interface available during any given trip. Notable information displayed by Figure 33 is the several individual polylines that compose a full journey, and the user's location and the highlighted current polyline representing the user's current step. Figure 34, highlighting the journey information component, requires a more detailed explanation. Numbered areas represent:

- (1) The current step the user is on, compared to the total number of steps.
- (2) A line representing the whole trip, divided between the individual polylines representing each step in the journey. Additionally, a dynamic user icon is displayed to indicate the user's precise position in relation to the whole journey.
- (3) A text box intended to draw the user's attention, containing a short message crafted to increase the user's probabilities of reaching the next step.
- (4) A "speaker" button that plays a longer version of the written message, allowing users to keep informed and on track through different sources.

3.4 LLM Prompts

As previously demonstrated, LLM calls are used extensively and frequently throughout the system. For every call the system performs, a query or input text is provided to the LLM to generate the intended response, these queries are called prompts. Prompt design significantly impacts the quality of the output, therefore, it is imperative to write these inputs with purpose. Prompt engineering is the process of designing, structuring and refining prompts, text inputs or instructions provided to LLMs in order to guide them into generating more cohesive and relevant responses (IBM, 2025a). Whether attempting to synthesise journey steps, generate failure preventions, orchestrate tool usage, or simply handle incoming user messages, appropriate and efficient prompting is crucial to guarantee the desired outcome.

3.4.1 GPT Prompts

The vast majority of LLM prompting within this system is directed at OpenAI's GPT models, following the few-shot, role assignment prompts. These specify exactly how the LLM should behave, their assumed identity, the expected input and output, and a detailed, structured explanation of its operational logic. Following OpenAI's API documentation guidelines, these instructions are distinctly structured (e.g., utilising distinct system and user messages) to optimise model adherence.

Additionally, a critical component of each call, meant to accompany the prompt, is the enforcement of a typed output structure. This guardrail transforms LLMs from unpredictable tools into reliable, deterministic data providers, safely allowing for pipeline integration. Structured output can be described and enforced through strictly typed schemas via the Pydantic Python library.

While prompt engineering is needed throughout the system, an important distinction should be drawn when analysing the intent behind each LLM call. Understanding how to handle agentic workflow's system prompts in contrast to singular function calls is crucial. This distinction is not rooted in the complexity of the prompt, but in understanding the primary goals and characteristics of each one. A functional LLM call's goal is to map a specific input directly to a specific output in a single inference step. They focus solely on immediate single task completion, providing exact formatting rules, specific few-shot examples of inputs and outputs, and clear boundaries to prevent hallucinations.

```
  instructions = f"""
     You are a "Resilient Route Architect" for an elderly-focused travel app.
     Your goal is to format incoming data into a graph node in a way that fits the existing travel graph
nodes.

     ### INPUT:
     A list of Google Maps trip steps and a specific failure situation in natural language.

     ### OUTPUT OBJECTIVES:
     An array of risk objects

     ### RULES FOR RISK FORMATTING:
         1. Summarise the natural language input into the core failure mode using fewer than 5 words
(e.g., 'Missed Bus 32', 'Panic Attack', 'Trip and Fall').
         2. The "label" must give an explanation of how the failure state was reached (e.g. "Uneven
pavement hazard", "Missed Right Turn"). Avoid using the "node_to" as a "label".
         3. Keep all labels and descriptions concise for elderly users.
       """
```

Figure 35 An example of a functional LLM call in charge of producing an outcome based on a given input

On the other hand, in a workflow with an increased degree of agency, the LLM takes the role of a reasoning engine driving a loop. The system prompt has the goal of defining a persona, rules of engagement and guiding the model through an iterative reasoning and acting loop (previously referred to as a ReAct loop). This prepares the model to receive inputs and, based on the available context, decide if it requires the use of external tools, evaluate the results of the used tool, and deem the task completed.

3.4.2 Anticip8 Prompts

Prompting Anticip8 is a considerably more time-consuming task, given the fact that it requires specific context, profile and call trigger crafting. Adapting Anticip8's needs in order to fit into a working data pipeline required a significant amount of trial and error, culminating in dedicated logic to address the natural language processing caused by the lack of structured output support. An additional complication encountered when working with the Anticip8 API is frequent throttle issues. A pipeline like the one Anticip8 is implemented in requires several consecutive API calls. Unfortunately, when trying to perform three to four consecutive calls, the API times out calls for a period ranging from 10 to 60 seconds, causing significant delays to an already time-consuming process. To tackle these timeouts, an additional backoff function was implemented to catch and retry API failed interactions.

Anticip8 has two main use cases, generating and ranking actions or only ranking the most probable next action without any new generated ones. Both of these are useful for the different functions in which Anticip8 calls are made. To fit Anticip8 into the Journey Planner pipeline, two dedicated functions are employed. While they have different purposes, the overarching structure is the same.

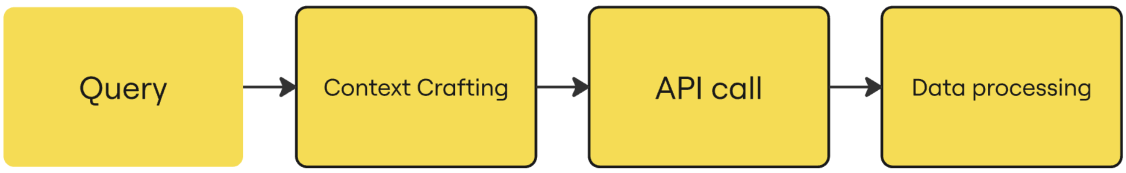


Figure 36 Anticip8 API call function data flow

This is a three-step process where the initial step dynamically crafts the relevant information required to compose a valid Anticip8 API call. This is an important step, since the API rejects calls that contain "insufficient" or "irrelevant" content. When crafting the call's context, a key property that allows us to bias Anticip8's output is the “subject”. Subjects like *"Negative situations for user's journey"* seem to drastically change Anticip8’s approach compared to other prompting modifications, such as inserting more relevant information in the *“recent_history_text”* or the *"triggers_text"* properties. This is, even if significantly negative recent history or triggers are given, the system will prioritise contextual instructions explicit in the subject. After relevant context is given, the Anticip8 call is made, returning a data structure containing information that includes an array with the provided or generated ranked actions based on a probability property. Since the API requires Anticip8’s rankings to include a minimum of a single generated action, when trying to only rank provided actions, filtering is required. It is important to note that oftentimes, Anticip8’s generated actions are ranked at the top, interfering with objectively more appropriate organic rankings. Finally, based on the function’s purpose, the returned data structure gets processed into a more fitting structured object for the pipeline.

3.5 Conclusion

As explained throughout the chapter, SafeStep employs diverse techniques to handle heavy processes required by AI while maintaining a smooth user experience focused on supporting non-tech-savvy users. By leveraging an asynchronous backend, the system bridges the gap between complex data pipelines and an accessible frontend. Furthermore, while the project presents a working prototype, it is currently on its first iteration, committed to continuously acknowledging bugs and edge cases, and implementing new features.

4. Testing, Experimentation and Comparative Evaluation Outcomes

4.1 Introduction

This chapter experiments with and explores the diverse methods of generating failure points that can arise during the users' journeys and the potential impact of tailored interventions to keep users on the correct path. For this, two testing phases will be in place. First, the core application's pipeline output testing, complex data in the form of a directed graph, will be analysed with help from the visualisation feature. The purpose of this testing is to examine the quality of the generated interventions and probability recalculation, and to identify the best-performing model combination to be implemented during the second testing phase. Finally, once the best model combination is identified, a live testing phase will begin, during which users will be taking the mobile application with them during some of their public transport journeys throughout a predefined period.

4.2 Graph Generation Testing

4.2.1 Methodology

To streamline this exploratory step, this project uses the previously mentioned Journey Planner visualisation tool in the browser. Before initialising the testing, it is necessary to set the base profiles the models will take in as context. To properly represent the application's target demographic, there were two carefully crafted profiles, suffering from dementia or physical frailty.

Details	Profile 1	Profile 2
Identity		
Name	Rosa	Harold
Age	75	82
Living Status	Lives alone, semi-independent	Lives alone, receives part-time caregiver support
Clinical Context		
Condition	Early-stage Dementia	Physical Frailty Syndrome
Cognitive Load / Physical Load	High susceptibility to "Sundowning" (increased confusion in late afternoon)	Low physiological reserve, high fatigue after minimal exertion
Symptoms	• Short-term memory lapses • Spatial disorientation • Executive function decline • Sensory overstimulation	• Reduced muscle strength • Slow gait speed • Poor balance and fall risk • Low endurance • Delayed recovery after illness
Behavioral History		
Navigation Errors / Mobility Limitations	**Bus Accuracy:** 20% failure rate **Directional Logic:** 25% failure rate **Waypoint Memory:** 10% failure rate	**Walking Tolerance:** Needs rest after 5-10 mins **Stair Navigation:** Requires handrail (40% difficulty without) **Assistive Devices:** Uses a cane outdoors
Anxiety Triggers	Crowds, loud echoes, complex transit hubs, and rushing commuters.	Fear of falling, icy sidewalks, long standing times, rushed environments, and poorly lit spaces.

These profiles attempt to accurately represent common patterns in behaviour and symptoms that an individual suffering from early-stage dementia or Physical Frailty Syndrome faces. For the models to perform to their best abilities, an appropriate context has to be delivered. This not only includes relevant symptoms but navigational history or other health vulnerabilities that reflect how these symptoms progress. Furthermore, keeping a precise log of anxiety triggers can give the selected model the defining information to properly generate relevant failure scenarios and weigh interventions.

For each journey, there are four different combinations of Anticip8/GPT.

	Failure Node Generation	Probability Recalculation
Iteration 1	Anticip8	GPT-5
Iteration 2	Anticip8	Anticip8
Iteration 3	GPT-5	GPT-5
Iteration 4	GPT-5	Anticip8

The testing data was collected from running all permutations on three different routes of varying lengths. It is important to note that with each feature run the outcomes will change slightly, but should have a similar effect given that both failures and preventions are based on the user's profile.

	Origin	Destination
Route 1	Trolley Park, Brent Cross, London NW2 6GJ	51.5860469, -0.2071016 **(Testing coordinates as input)**
Route 2	Hendon Library, London NW4 4BQ	91C Church Road, NW4 4DP
Route 3	The Regent's Park, London **(Testing no postcode)**	Manor House, Friern Barnet Ln, London N20 0NL

The following is an example of the analysis and further comparative process employed to reach the results explained in the results section.

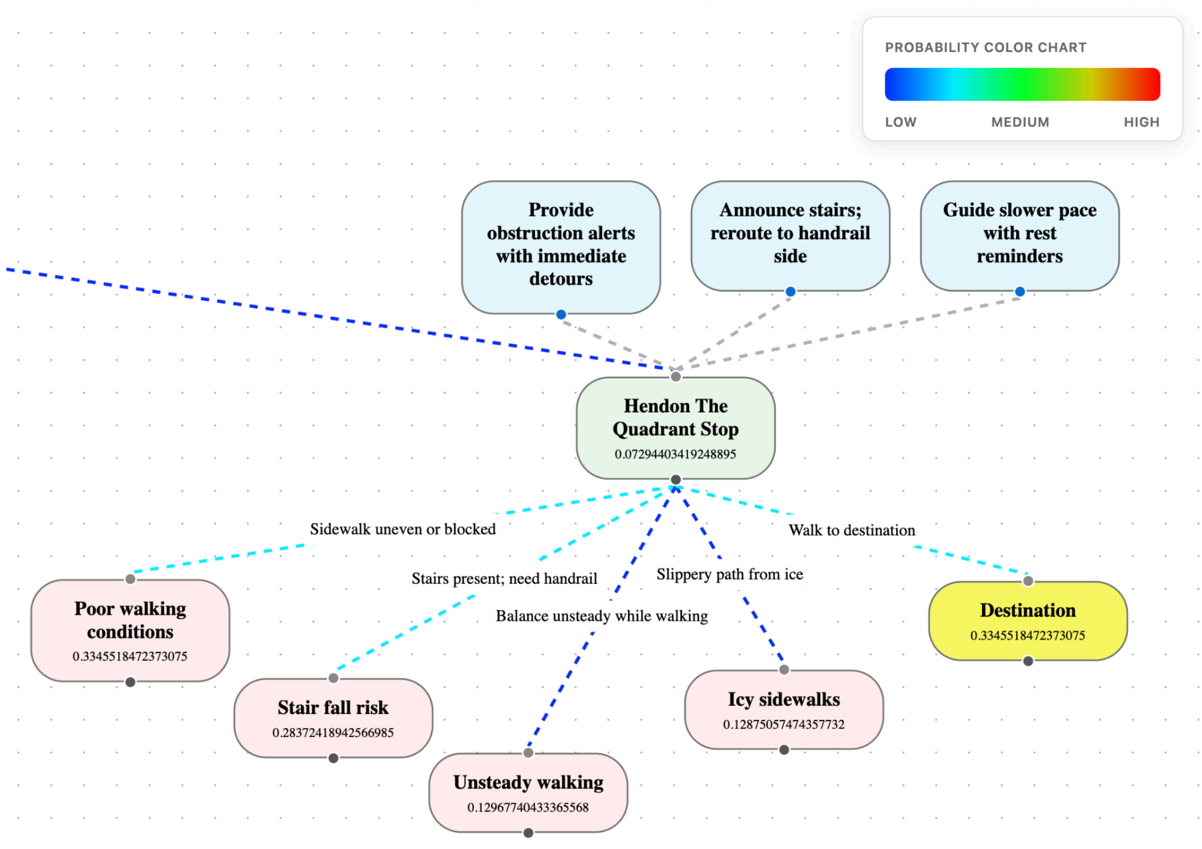


Figure 37 plain node with no interventions applied

The above screenshot shows the base state of a node generated using the Anticip8/Anticip8 combination (failure node generation, initial probability generation, and probability recalculation based on applied prevention done by Anticip8), using the frailty profile. The following screenshots will showcase how the different probabilities change when selecting each of the available preventions.

The initial chances are:

- Failure node 1: Poor walking conditions due to uneven sidewalk - 0.3345
- Failure node 2: Stair fall risk - 0.2837
- Failure node 3: Unsteady walking due to own balancing issues - 0.1296
- Failure node 4: Icy sidewalks (fall risk) - 0.1287
- Successfully walking to destination - 0.3345

In the current base example, Anticip8 calculates that there are equal chances of the user safely walking to their destination or encountering poor walking conditions that prevent the user from continuing their journey.

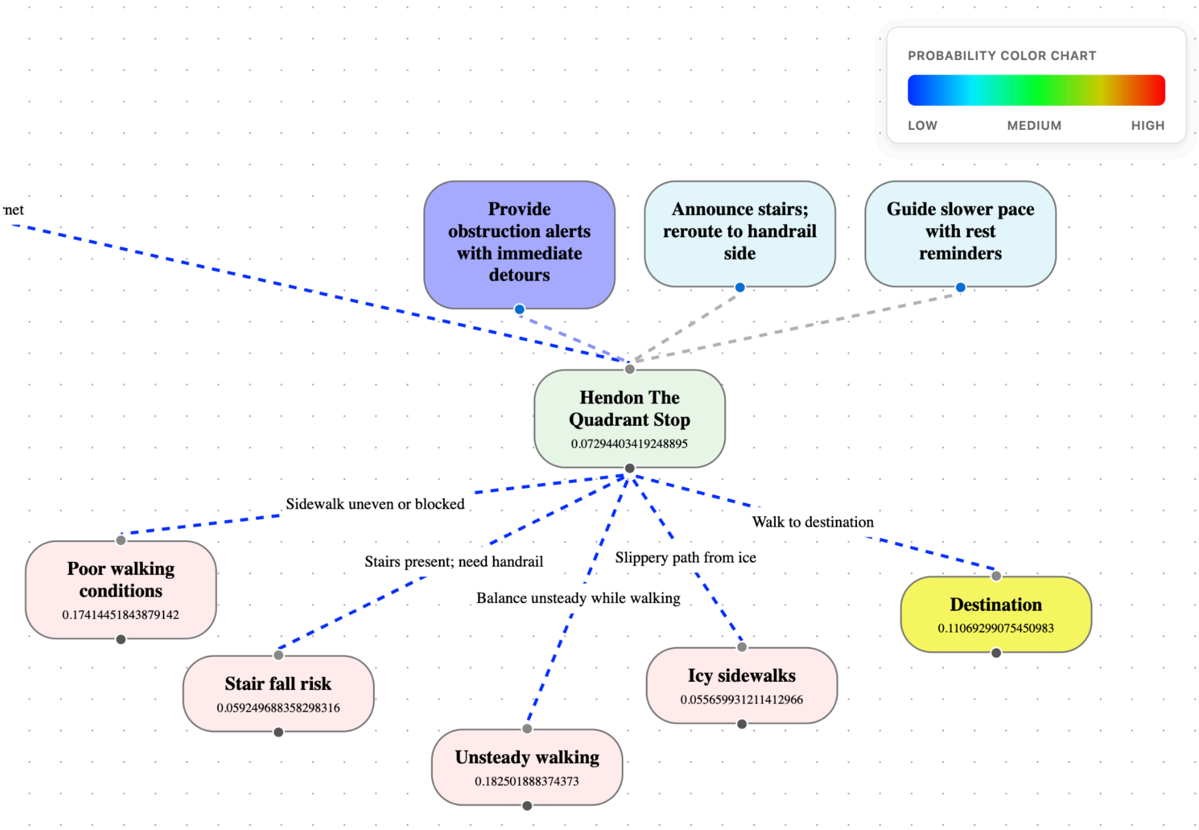


Figure 38 probability changes with the first intervention applied

The first intervention is intended to specifically tackle failure node 1, prompting the user with a voice command about obstructed pathways.

The current chances are:

- Failure node 1: Poor walking conditions due to uneven sidewalk - 0.1741
- Failure node 2: Stair fall risk - 0.0592
- Failure node 3: Unsteady walking due to own balancing issues - 0.1825
- Failure node 4: Icy sidewalks (fall risk) - 0.0556
- Successfully walking to destination - 0.1106

The observable changes are a general decrease in changes across most of the nodes, with the most significant changes on failure node 1, 2, and destination. On initial analysis, changes on failure node 1 make sense, but significant changes on failure node 2 and a decrease in walking to destination's success rate seem odd. Additionally, the model increased the probabilities of the user's capability of maintaining balance, which, given the nature of the prevention being providing easier to navigate pathways, does not align with the expected outcome.

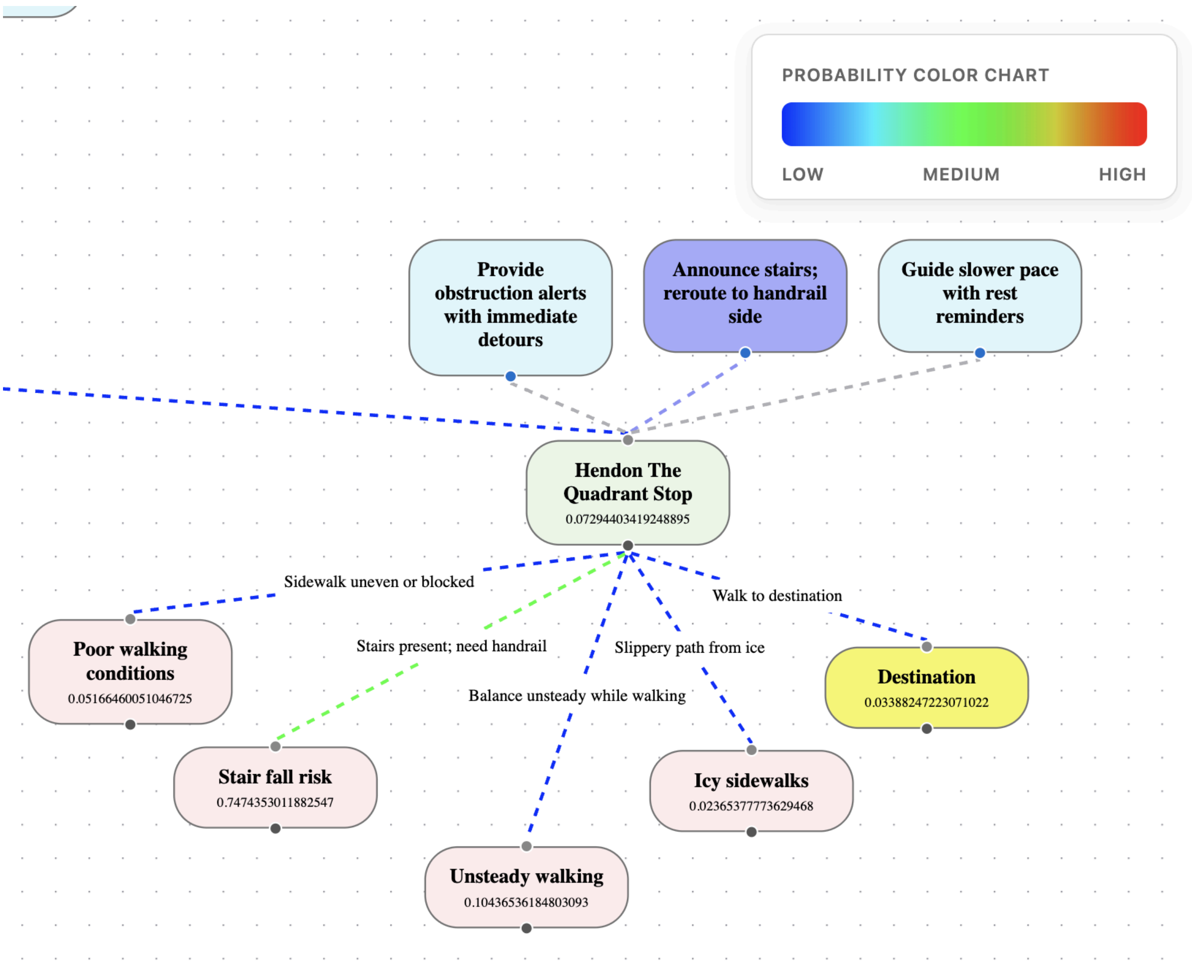


Figure 39 probability changes with second intervention applied

The second intervention is intended to specifically tackle failure node 2, prompting the user with a voice command about incoming stairs and advising them to use the handrail.

The current chances are:

- Failure node 1: Poor walking conditions due to uneven sidewalk - 0.0516
- Failure node 2: Stair fall risk - 0.7474
- Failure node 3: Unsteady walking due to own balancing issues - 0.1043
- Failure node 4: Icy sidewalks (fall risk) - 0.0236
- Successfully walking to destination - 0.0338

The observable changes reflect some decrease in failure nodes and a significant increase in failure 2 node's probabilities. On initial analysis, failure node 2's probabilities *should* significantly go down when this prevention is applied, but instead, there is a threefold increase. Additionally, this prevention has an impact on failure node 4 and destination success, both being decreased, which does not align with the expected outcome.

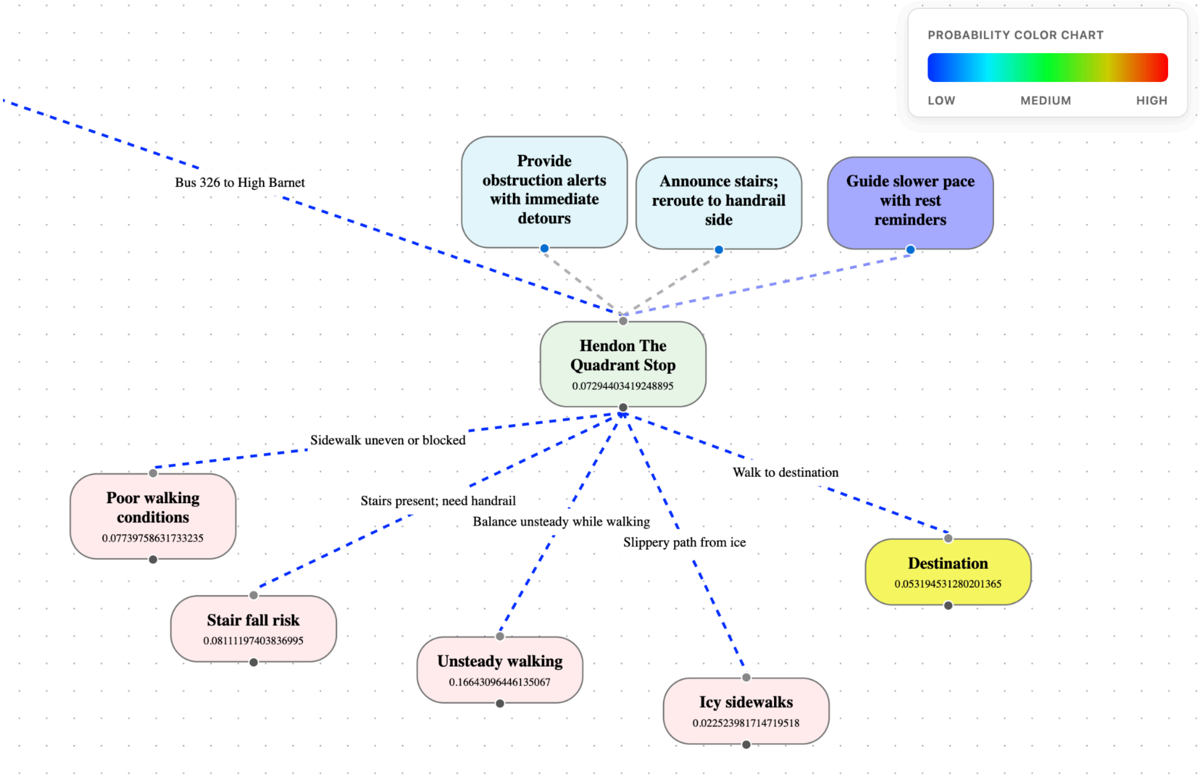


Figure 40 probability changes with third intervention applied

The last intervention is intended to aid the user's walking difficulty, prompting the user with a voice command, encouraging them to have a slow pace and take breaks.

The current chances are:

- Failure node 1: Poor walking conditions due to uneven sidewalk - 0.0773
- Failure node 2: Stair fall risk - 0.0811
- Failure node 3: Unsteady walking due to own balancing issues - 0.1664
- Failure node 4: Icy sidewalks (fall risk) - 0.0225
- Successfully walking to destination - 0.0531

The observable changes reflect general probability decrease across most nodes, with failure node 3's probabilities increasing slightly. On initial analysis, while a general decrease of issues related to mobility is expected, failure node 3's probability increase is out of the ordinary. Additionally, this prevention has an impact on destination success, being decreased, which does not align with the expected outcome.

4.2.2 Results

The aforementioned analysis was conducted on the four different model iterations with both profiles. The objective was to find the best-performing combination, which was set as the default combination for the live testing period. All model iteration's journey graphs can be found in Appendix A.

4.2.2.1 Frailty Profile

GPT-5/GPT-5

When selecting GPT-5 for both failure node generation and probability recalculation, the results tend to be highly generic, and probabilities can be relatively over optimistic. Since the previously mentioned Anticip8 API throttle is not present, this iteration is the fastest, completing the route shown in 6:15 minutes. Nevertheless, the failure nodes generated lack the depth Anticip8 brings to the picture, e.g., GPT-5 produces standard physical risks such as "trip on curb". Furthermore, GPT-5's probability generation capabilities can be unrealistically forgiving, pushing probabilities of success to 100% in some cases. In conclusion, this combination is fast, but it can lack the nuances necessary for this use case.

Anticip8/Anticip8

When selecting Anticip8 for both failure node generation and probability recalculation, the results show highly personalised risk node generation but bring light to a significant Anticip8 downside. When recalculating probabilities based on the prevention applied, the results tend to be quite inconsistent, often having no correct path success increase after applying preventions that specifically target specific or all failure nodes. Furthermore, success probabilities often decrease when applying preventions, even when these are highly relevant to both failure nodes and profiles. Additionally, the Anticip8 API throttle makes this combination take the longest, with the exemplified journey taking short of 10 minutes to run. On the other hand, Anticip8 very successfully integrates the user’s clinical profile in the generation process, resulting in highly tailored failure nodes.

GPT-5/Anticip8

When selecting GPT-5 for failure node generation and Anticip8 probability recalculation, it’s notable that this combination is the least synergistic, with results showing this combination inheriting the worst attributes of each model. The exemplified journey took 7:47 minutes to run, with GPT-5 initiating the process by generating generic failure nodes such as “trip and fall”, missing key aspects of the user’s symptoms or profile. Anticip8 then recalculates probabilities based on these non-tailored nodes, resulting in arbitrary and inconsistent calculations seen in the Anticip8/Anticip8 combination. This iteration fails to leverage Anticip8’s personalised node generation while suffering from its erratic probability calculation.

Anticip8/GPT

This iteration successfully balances clinical personalisation and better probability weighting, showing the most functional results. By employing Anticip8 for generation, the system captures user-specific risks, while leaving the probability recalculation to GPT-5, avoiding Anticip8’s inconsistent outcomes. Taking 8:05 minutes to run, this combination provides the most consistent, tailored, and rational output.

4.2.2.2 Dementia Profile

GPT-5/GPT-5

Similar to the frailty profile, having the GPT-5/GPT-5 combination results in fast but surface-level results. GPT-5 generates generic cognitive risks that fail to take into consideration the specific user’s triggers. On the other hand, this combination avoids Anticip8's characteristic API throttles and delivers the probability recalculation strengths GPT-5 carries.

Anticip8/Anticip8

When selecting Anticip8 for both failure node generation and probability recalculation for the dementia profile, results show an impressive failure node contextual generation that is significantly bottlenecked by slow run times and inconsistent probability recalculations. Due to the API throttling, this run took short of

10 minutes, a similar time to the previous Anticip8/Anticip8 run. However, Anticip8 successfully considers the user's clinical context, producing nuanced failure nodes based on Rosa's susceptibility to sundowning. Unfortunately, Anticip8's probability recalculation performance remains behind compared to GPT-5s, with the system often triggering arbitrary drops in success probabilities.

GPT-5/Anticip8

Similarly to the frailty profile, when employing the GPT-5/Anticip8 combination, results fall flat by mostly ignoring the complex dementia cognitive profile when generating failure nodes while introducing probability recalculation inconsistencies. While finishing the task in a relatively short time, results show generic failure nodes, forcing Anticip8 to produce unpredictable probability recalculations, failing to reflect the actual benefit of the applied interventions.

Anticip8/GPT-5

This combination of Anticip8 generating failure nodes and GPT-5 recalculating probabilities, once again shows to be the optimal pairing, providing user-specific clinical context and similarly efficient recalculation outcomes. Anticip8 accurately reflects the user's environmental vulnerabilities, generating relevant nodes such as "Transit Hub Confusion" and "Sensory Overload". Subsequently, GPT-5 recalculates probabilities with expected accuracy. This iteration delivers improvements without the flaws carried by both models.

After careful analysis, the Anticip8/GPT-5 is clearly the best performing combination. This model iteration puts forward the best of both models, leveraging Anticip8's personalised failure node generation and GPT-5's consistent, though sometimes optimistic, probability recalculation. With a selected winner, the live testing model combination can be set, and SafeStep can move to the next testing phase.

4.3 Live Testing

4.3.1 Methodology

For the second testing method, seven users were asked to download the mobile application and use it in some of their daily journeys throughout the period of a week. While users were not part of the target demographic, they were provided with

SafeStep accounts set up with the two aforementioned profiles. Furthermore, users were prompted to read their profile description and keep it in mind while interacting with the application. The purpose of non-simulated testing is to test the system in multiple different ways that would not be possible otherwise. To its strength, they would be testing the SafeStep application's general functionality and its resilience in a real-world environment. Then, this testing would challenge how useful the AI-generated preventions are and how well the application implements measures to make sure users are aware of them. Finally, it would test the user interface's accessibility features such as font size and weight, clarity and volume of audio alerts, and how intuitive the user interface is to navigate.

After the testing period concluded, users completed a questionnaire describing their experience using the application. The questionnaire responses and application instrumentation-derived insights were used to infer testing results.

4.3.2 Results

Based on trip history, participants have used SafeStep to calculate and perform their public transport journeys 26 times in the span of a week. Participants used the app as many as five times and as few as three times during this period. Furthermore, instrumentation metrics show that users got off-track 10 times throughout their journeys.

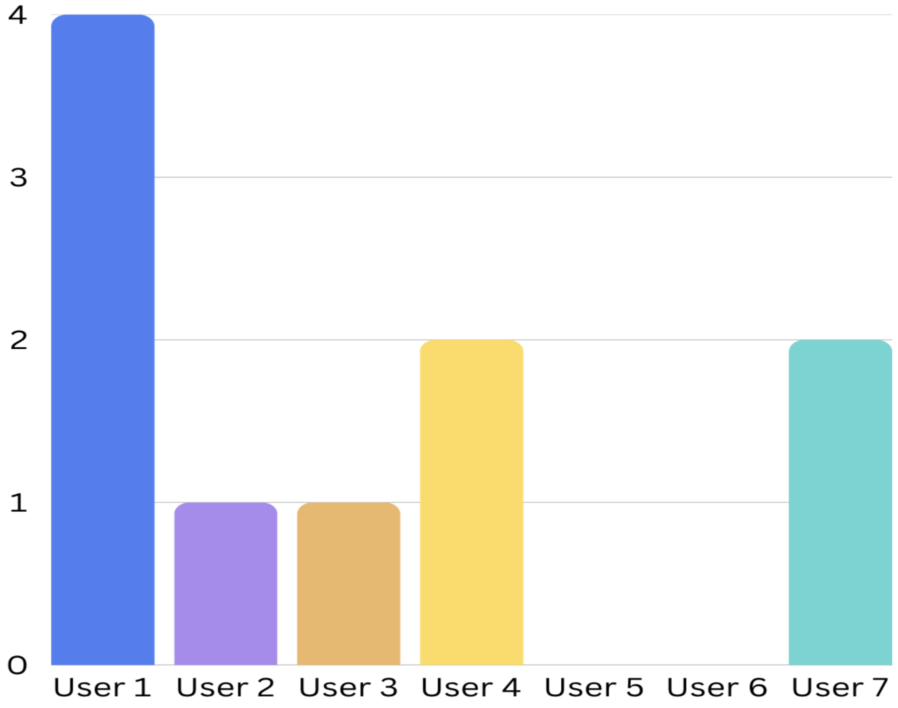


Figure 41, Amount of times SafeStep deemed the user got lost

Additionally, all users reported that when they got off track, the system recognised this “immediately” or “fairly quickly” (In the first 30 seconds of going off track). This information, paired with intervention feedback, can lead to concluding that interventions have a positive impact on users' trips.

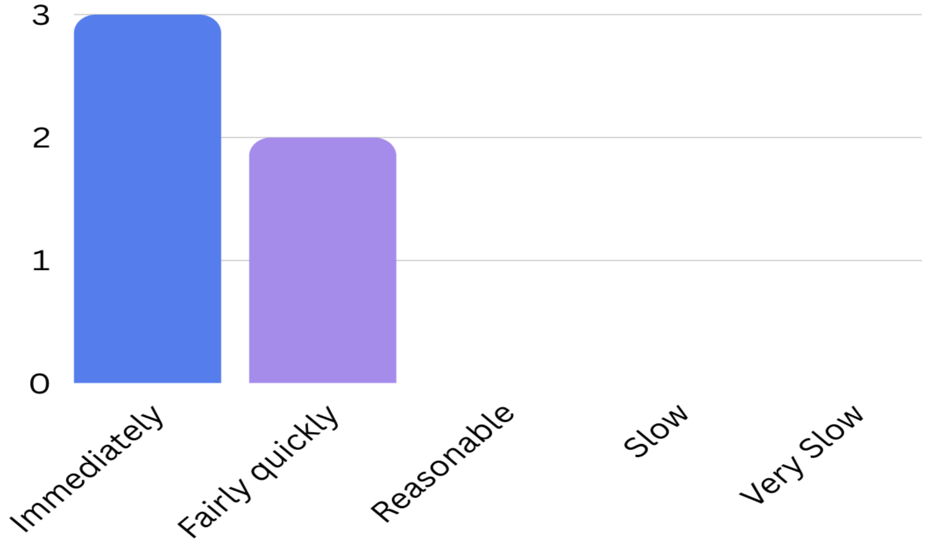


Figure 42, How quickly did the system detect deviations

4.3.2.1 User Interface

When developing an application for this specific demographic, having a user interface that reflects their needs is critical. The questionnaire collects relevant user information through enquiring about text adequacy, route visual elements not risking overwhelming the user and general information displayed in the interface being useful and easy to understand. User answers will have a direct impact on the application's design and will be taken into consideration for future design iterations.

The user's answers draw a clear picture of the application interface's strengths and weaknesses. On one hand, the majority of users rated font size, colour and contrast on the lower side of the Likert scale.

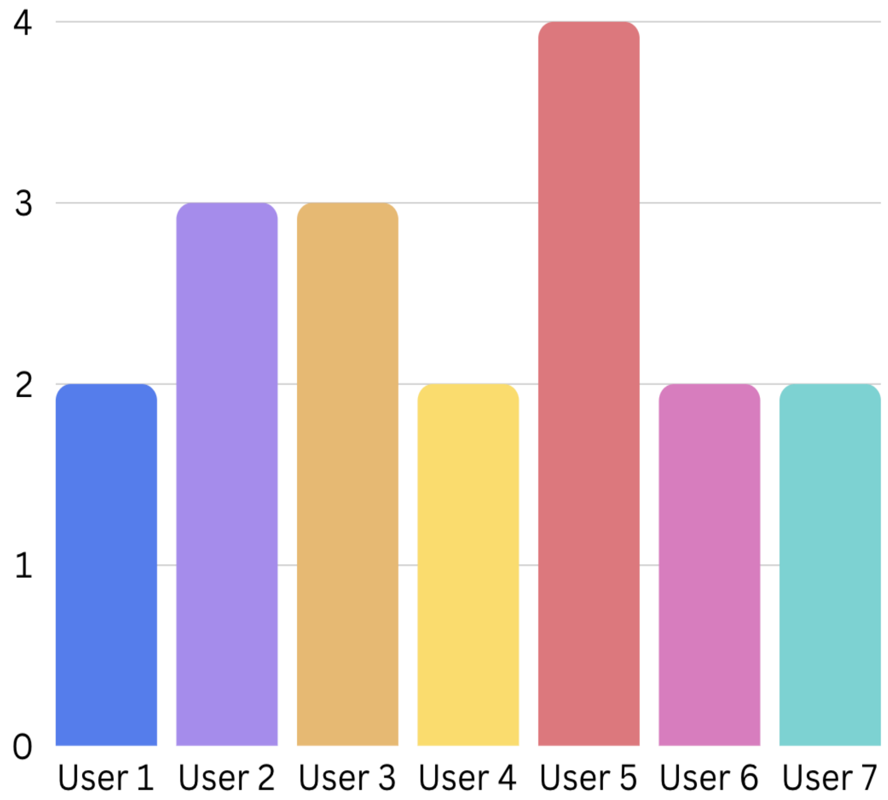


Figure 43, Question 1 - Text size colours and contrast were appropriate

Given that the older demographic tends to significantly struggle with font size, this is a metric that will be strongly taken into consideration when redesigning the application's interface. On the other hand, users evaluated the route visual elements, such as polylines and icons, as simple and not overwhelming.

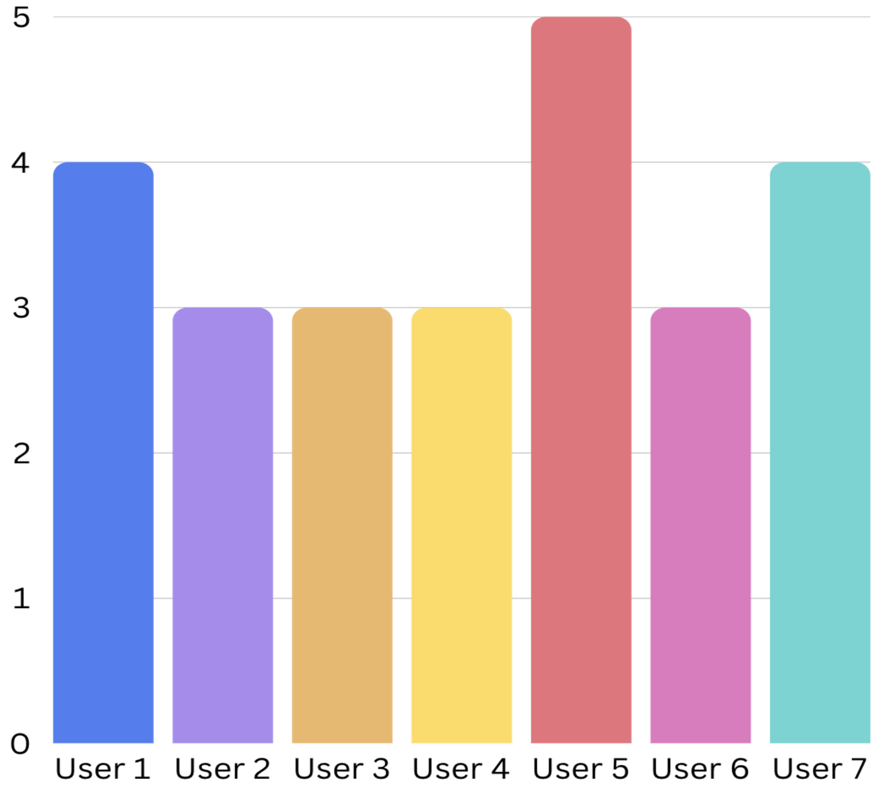


Figure 44, Question 2 - Route visual elements were simple and not overwhelming

Furthermore, other information displayed in the UI, such as current step number, current step prevention and users' relative positioning, was useful and easy to understand. This last metric was the most positively rated, with four users rating it with a 5 out of 5.

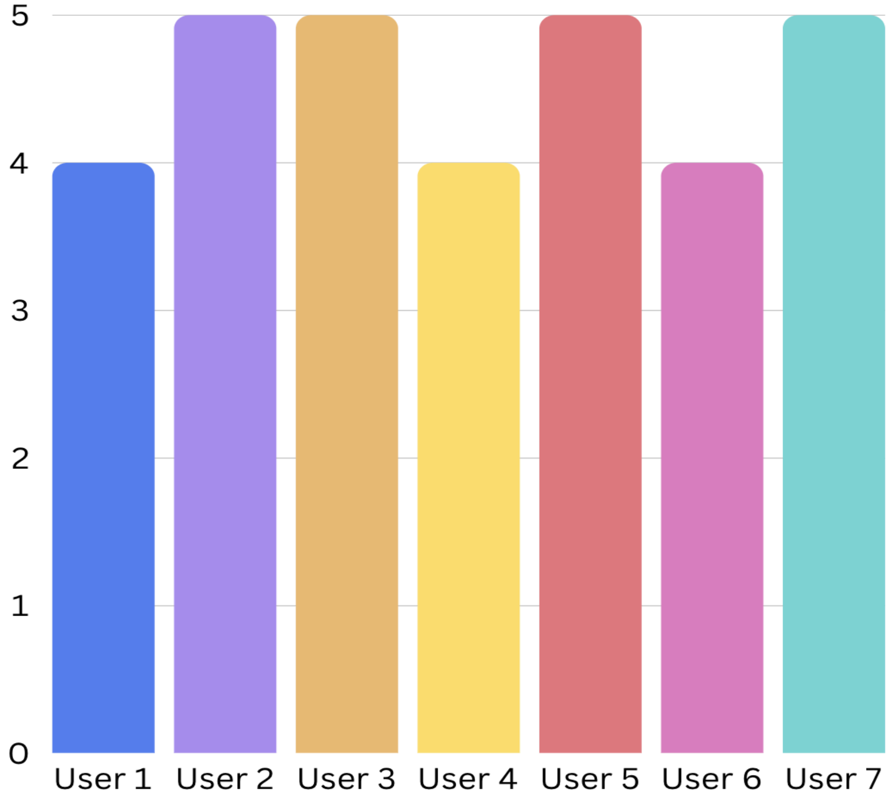


Figure 45, Question 3 - Information displayed in the UI was useful and easy to understand

4.3.2.2 Audio Intervention Quality

An instrumentation metric used to cross-check the self-reported usefulness of audio interventions is the "prevention_audio_plays" feature in the trip_history database table. This feature shows how many times users pressed the button to play the audio prevention feature on each trip. This is an insightful metric because it allows for data plotting throughout all trips, showing a more encompassing perspective.

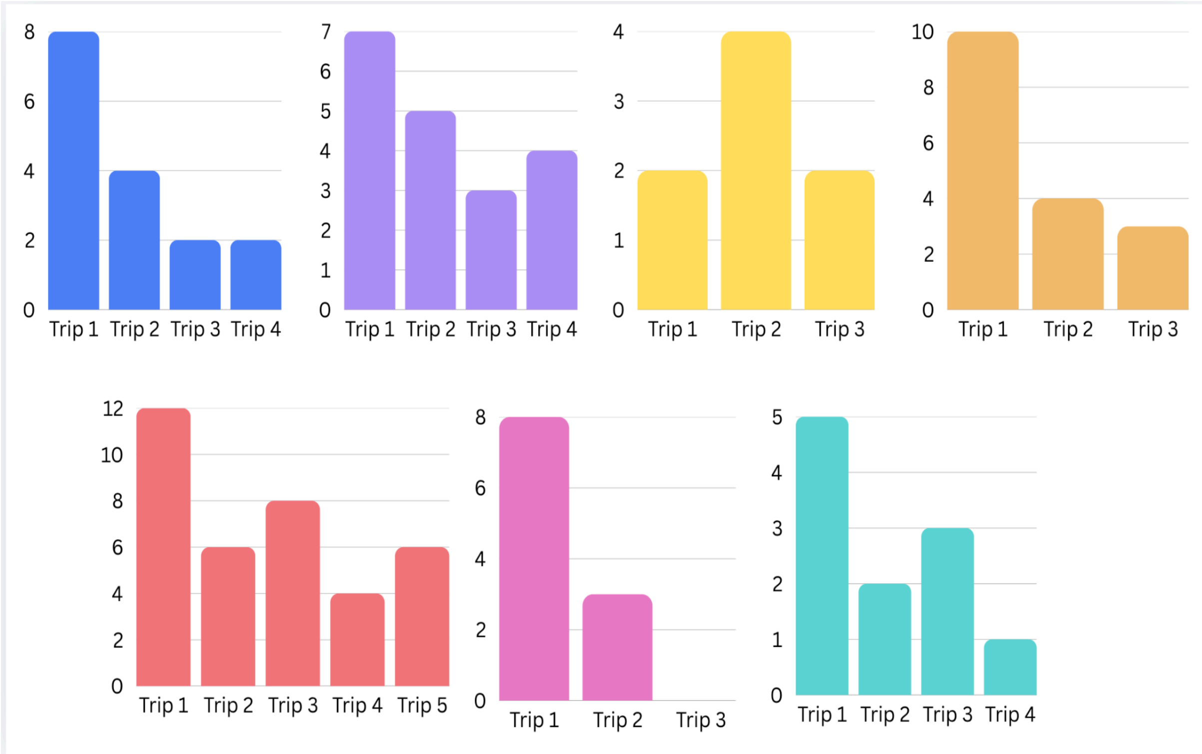


Figure 46, Amount of times users used the Audio prevention feature each trip user 1-7 left to right

When looking at the graphed data, not only is it observable that users show an interest in the feature during their first trips, but it could also be interpreted as their progressive loss of interest in it. On the other hand, when cross-referencing these data with the derail data, it is observable that users 5 and 6 had no derails and consistently triggered audio preventions. This potentially highlights the feature's value. When users were asked about this feature through their questionnaires, the objective of the questions was to both denote how the demeanour of the selected voice was and how closely related the text version of the interventions was to the audio counterpart.

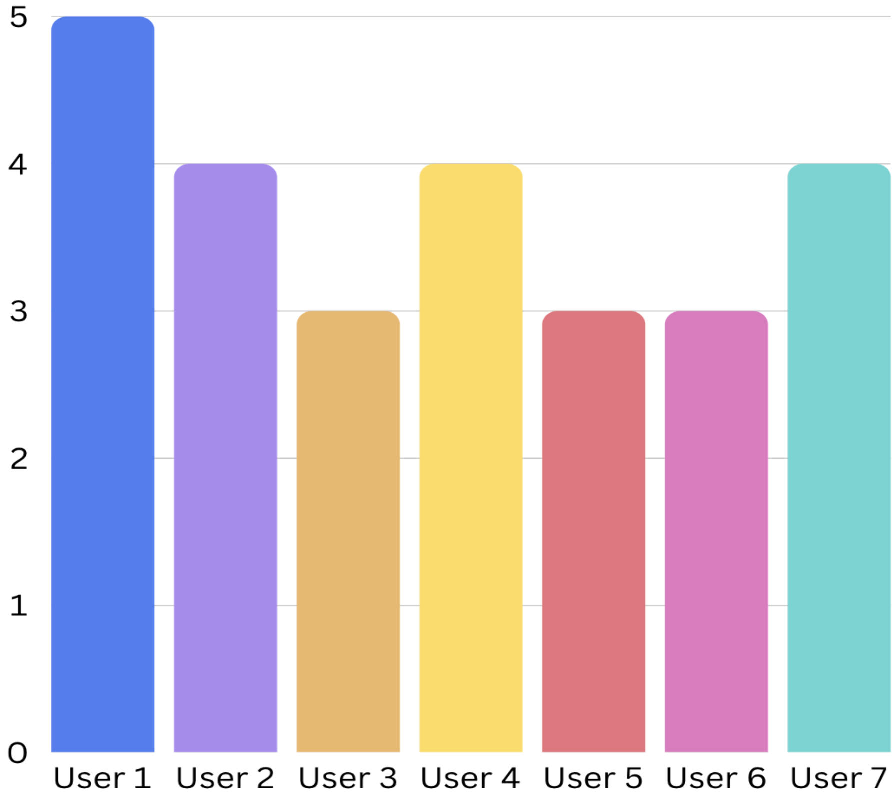


Figure 47, Question 4 - Voice instructions are accurate representations of text displayed

Users strongly agreed about the tone and general demeanour of the voice being calm and non-robotic sounding, a crucial quality when dealing with potentially distressed users. Additionally, users reported a strong similarity between text and voice interventions. This is important, since text is meant to be a compact, quick burst of information, while the audio form aims to be more verbose and slow-paced. Finally, users reported audio features, including interventions and step narration to cue on time and audio volume to be adequate even in loud environments.

Additionally, general intervention feedback leads to the conclusion that interventions felt reassuring and non-panic-inducing.

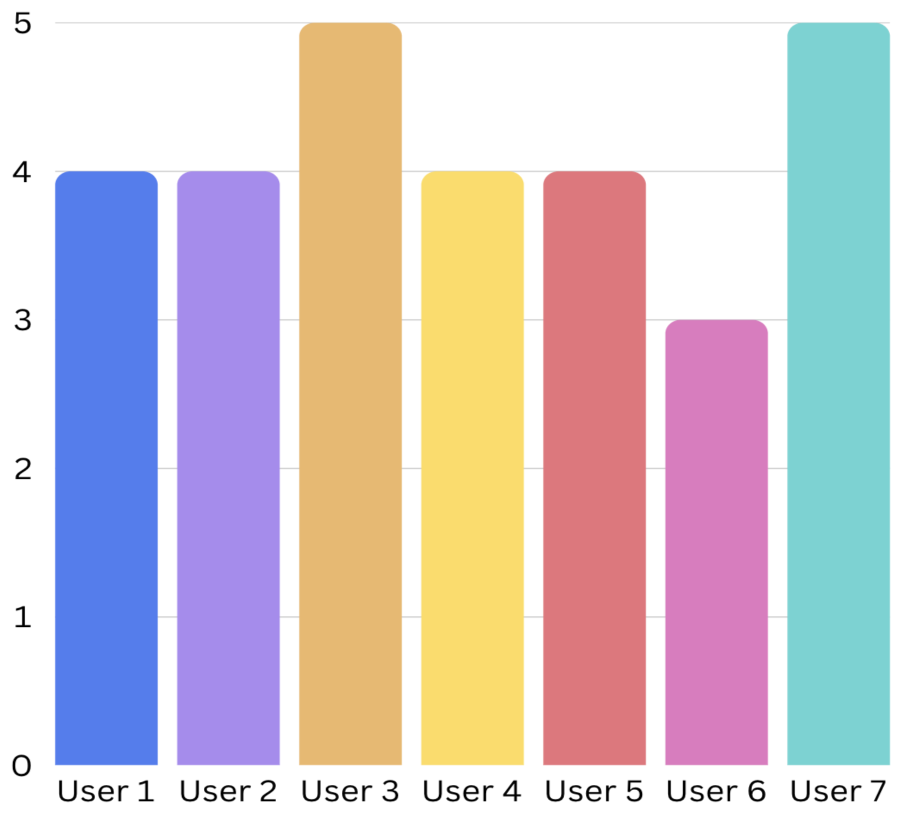


Figure 48, Question 6 - interventions felt reassuring rather than panic-inducing.

Furthermore, the majority of users reported interventions to be generally appropriate for the intended demographic, describing them as being “Very helpful” and “Properly worded and paced”. Despite this, one user mentioned interventions were not adequate for the step they were in and emphasised the intervention’s lack of relevance.

4.3.4 Conclusion

Given the allotted period and amount of trips, the live testing phase brought some valuable insights that solidified some of the system’s features while providing constructive criticism that highlighted areas of improvement. Based on this feedback, future system iterations will have adjusted text size and contrast to match the positive UI features, bringing the overall design to its best possible standard. Additionally, further modifications might include model prompting rewording, aiming to make interventions more intentional and relevant to the user’s situation.

5. Conclusion

5.1 Discussion

It has been established that the UK's elderly population faces both cognitive and physical impairments that hinder their independence and cost the UK government millions per year. While there are some applications that try to proactively help their elderly users, none are centred around journey planning through convoluted public transport systems such as TfL. This project attempted to, through the use of an innovative predictive technology delivered by Fountech AI, build a solution that allows for an alternative to traditional journey planning applications, keeping the user and their needs at the forefront.

Through multiple data pipelines and AI agentic workflows, SafeStep delivers a way for users to delegate appointment management, journey planning and trip safety to a caring companion. Alongside this FastAPI-built system, SafeStep counts with both web and mobile interfaces, allowing users to test its journey generation capabilities. A visualisation tool is available on the web portal, allowing users to test the journey generation capabilities, displayed in an interactive directed graph form that allows them to understand potential trip failure scenarios and how the system would help to avoid them. Subsequently, SafeStep has a mobile version available that allows the user to take SafeStep with them through their journeys, taking advantage of their tailored safety interventions.

Live user testing was conducted during the week of 13th to 24th of March, with the help of seven users. Users were instructed to take the mobile application through their journeys, testing the journey-generated recommendations and paying attention to design details such as interface relevancy and voice command demeanour. After the testing period, users were asked to complete a simple questionnaire going over significant aspects of the application. Additionally, users were given access to a feedback feature, allowing them to flag any issues they encountered.

Testing results show the system is capable of tracking users through their journey in real time, identifying current location in relation to their journey and off-track deviations in an appropriate amount of time. Furthermore, users found intervention quality to be appropriate for their profiles, and voice commands to be suitable for elderly users, providing a calming, well-paced demeanour. On the other hand, while the main functionality, map visualisation, and interventions successfully provided a good user experience during journeys, feedback also highlighted areas of improvement. The mobile user interface's accessibility, particularly font size, colour and contrast, called for a significant redesign to better accommodate the target elderly demographic.

To conclude the discussion, Nicole Kairinos, Anticip8's product manager, reviewed SafeStep and provided an industry overview that can be found in Appendix C.

5.2 Limitations & Further Work

Due to the size of the project and how many technologies it implements, naturally, the initial plan and shape had to be rethought and recontextualised several times as the project developed to be what it is now.

Learning new technologies and integrating them into a system does not come without hiccups. While building agentic workflows was not new, communication between multiple of them was. Figuring out the best way to do so and understanding how to craft a FIPA-ACL package efficiently was challenging. Additionally, in the realm of these workflows, it is relevant to mention that they were the first thing built, and originally, they were going to have a significantly larger amount of participation in the journey planning life-cycle. Later in development, understanding the needs, limitations and time constraints, fitting Anticip8 into the project represented, the system had to turn away from the original relevance these workflows had. Finally, significant complications appeared once tackling the system's communication layer. Handling heavy, time-consuming processes such as journey planning with regular HTTP request/response brought several complications in practice. When searching for alternatives, a communication protocol that could simply flag a process as pending and deliver a response when it's finished, without the overhead associated with freezing the client, seemed ideal. Websockets' full-duplex connection allowed this independent communication, but also brought a fair amount of complexity. Statefulness, reconnection and even the smallest amount of scalability represented a trial and error process that overwhelmed a significant portion of the available time.

Having this in mind, these are the most significant areas of improvement, such as:

- **Websocket reconnection strategy:** The current websocket architecture attempts reconnection indefinitely every three seconds. This strategy is appropriate for a working/testing prototype, but it is far from ideal. In a production environment, it would introduce issues when faced with simultaneous mass disconnections. To solve this issue, a relevant fix would include performing a backoff similar to the Anticip8 throttle backoff, where the system increases time between reconnection attempts, eventually capping this time or closing the connection for good.

- **Websocket queuing strategy:** The current websocket architecture drops messages if there is not a fully working connection. This can create major communication issues and can be fixed by keeping undelivered messages from either side in a queue, only being delivered once a stable connection is established. Similar to the above websocket issue, the current approach to websocket messages is appropriate for the project's current state, but would benefit from upgrading.
- **Improved Anticip8 Prompting:** While the current prompting strategy successfully delivers the intended results and an optimal model combination has been reached, this does not mean there is not room for improvement. Fountech AI has been asked for feedback on how to improve SafeStep's prompting, and adequate changes will be made to achieve the best possible results.
- **RAG scaffolding integration:** The current email ingestion pipeline takes the necessary steps to support RAG functionality. This feature, meant to solidify the presence of the "companion" aspect of the app, would be extremely helpful to keep users in touch and up to date with their upcoming appointment's details.
- **Handling off-tracking corrections:** Once again, due to the complexity of the project, this is an area that was originally planned for but could not be integrated into the final application. When the system detects the user going off track, it recalculates the route from the point they deviated to their original destination. Originally, the system would have access to an array of different tools that would trigger based on the user's profile and configuration, further leveraging Anticip8's capabilities. While this feature integration is not complicated, given the current tool-based agentic architecture, it was clear that a reactive system like SafeStep should focus its efforts primarily on interventions aimed at preventing derails rather than corrections.
- **Dashboard Integration:** The current home page displays key information about the pre-constructed dementia or frailty profile assigned to the account. When envisioning the project's features and UI, the dashboard would display users'

appointment information, including system intersections such as rescheduling and cancellations and other useful features for both users and caregivers, such as reminders and device vitals. Due to time constraints and the testing phase not including the email/calendar pipeline functionality, these features were postponed to further project iterations.

To conclude, the development of SafeStep proved that there is a place for AI processes in systems designed for vulnerable populations. By proactively mitigating the risks that may arise during a commute, SafeStep reinstalls a primary degree of independence in the elderly and cognitively impaired users. SafeStep serves as a proof of concept that cutting-edge technology like Anticip8 can be leveraged to mitigate risks throughout countless fields, enabling AI to aid in use cases it has not yet been considered to.

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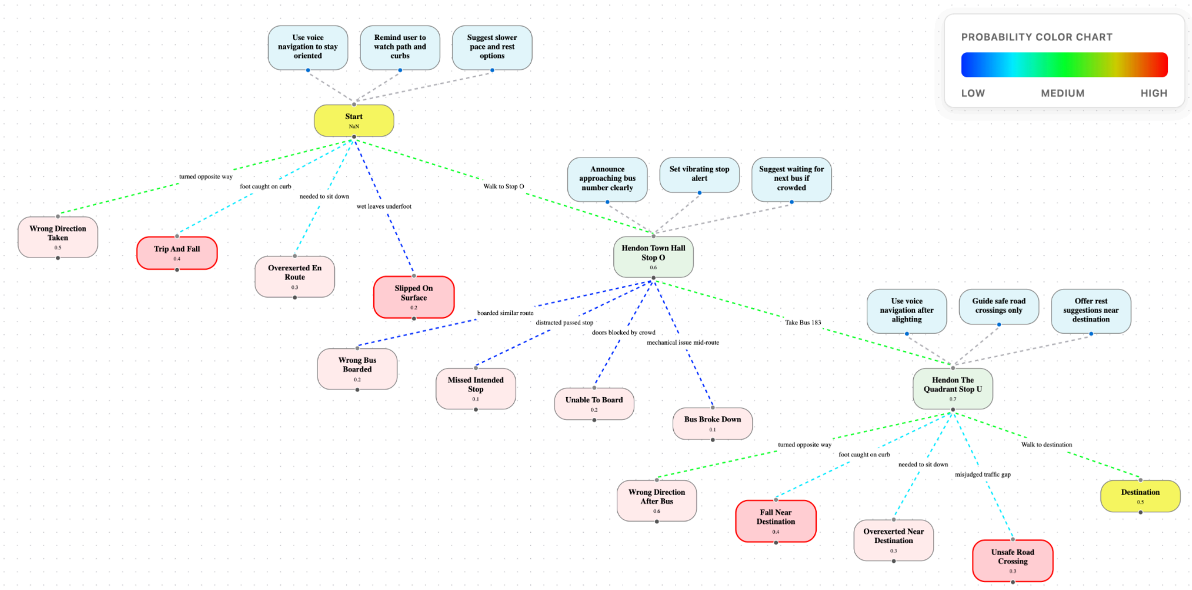
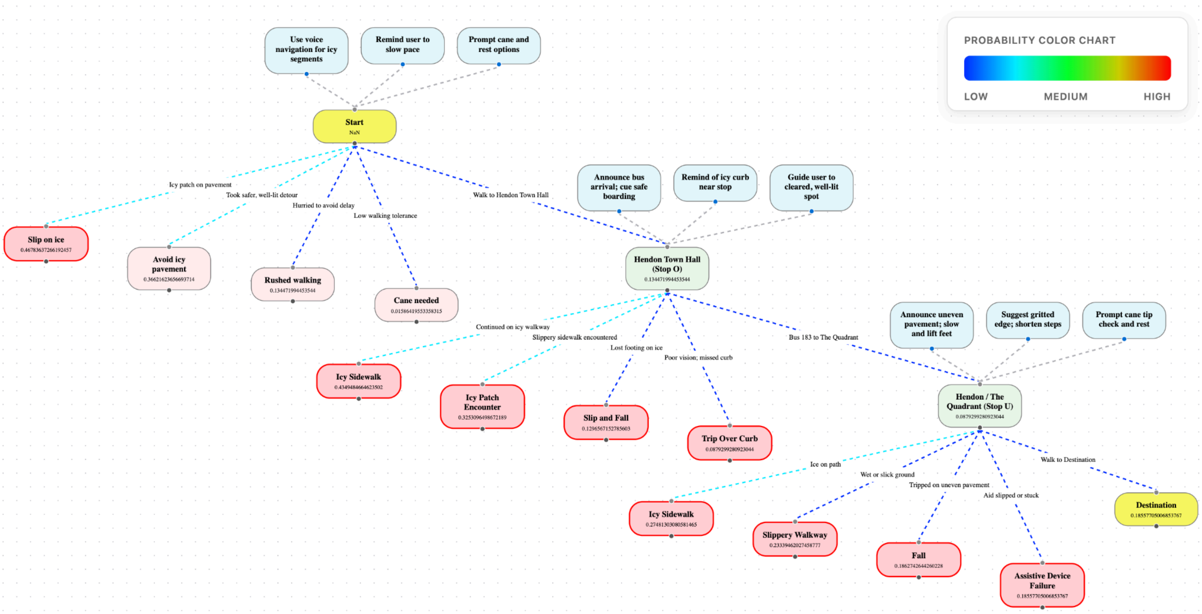
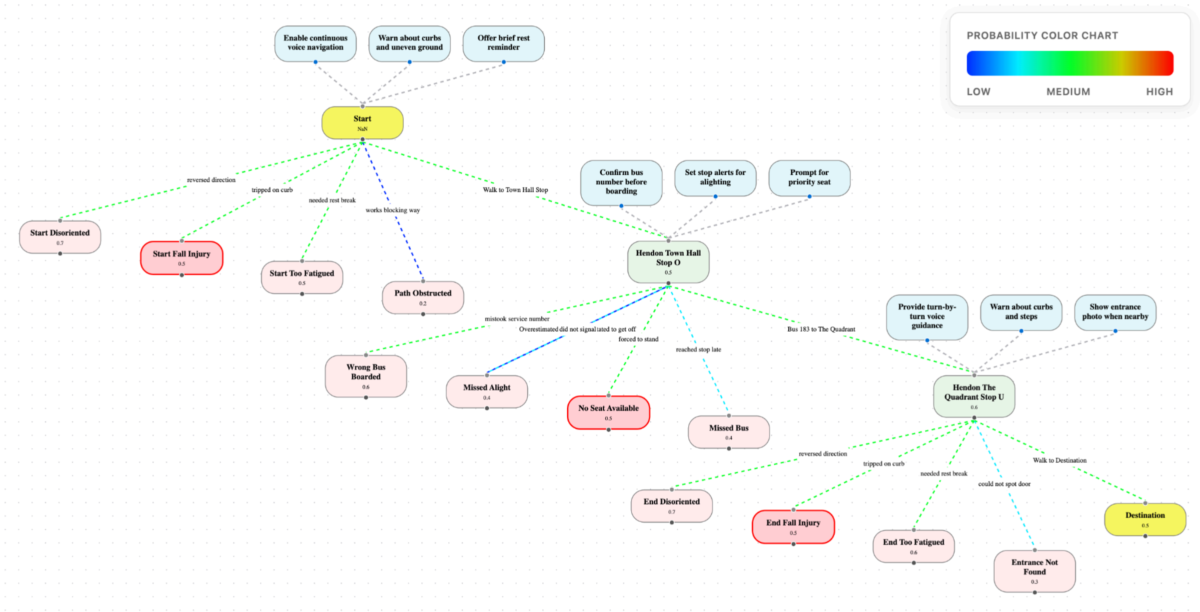
Appendix A - Journey Graphs



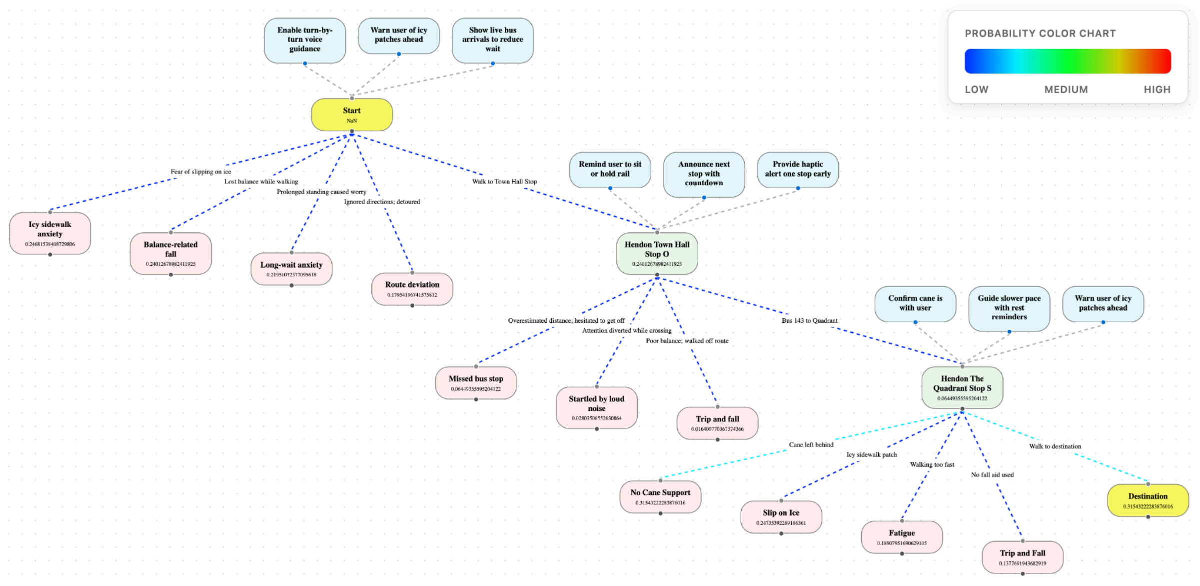
Journey graph - Frailty - GPT-5/GPT-5



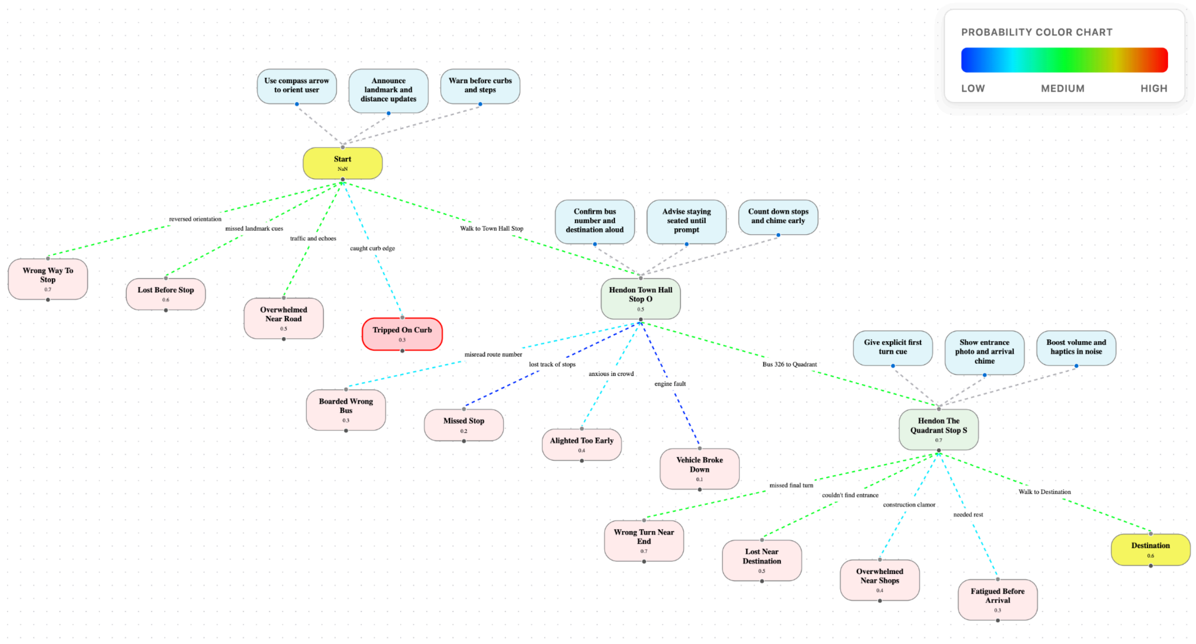
Journey graph - Frailty - Anticip8/Anticip8



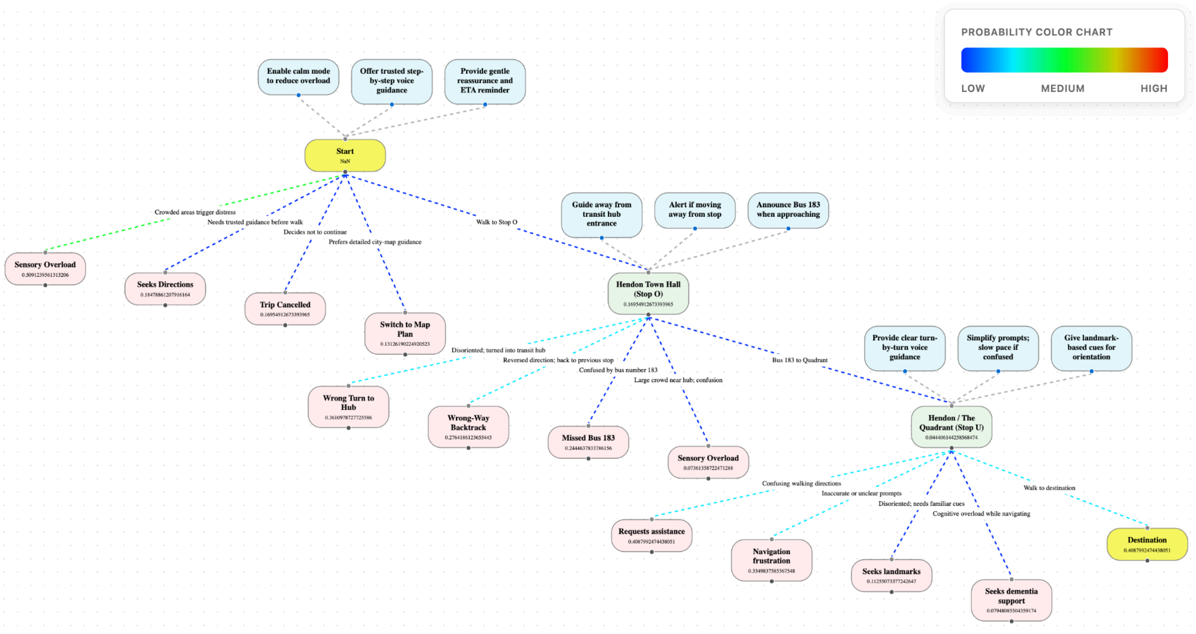
Journey graph - Frailty - GPT-5/Anticip8



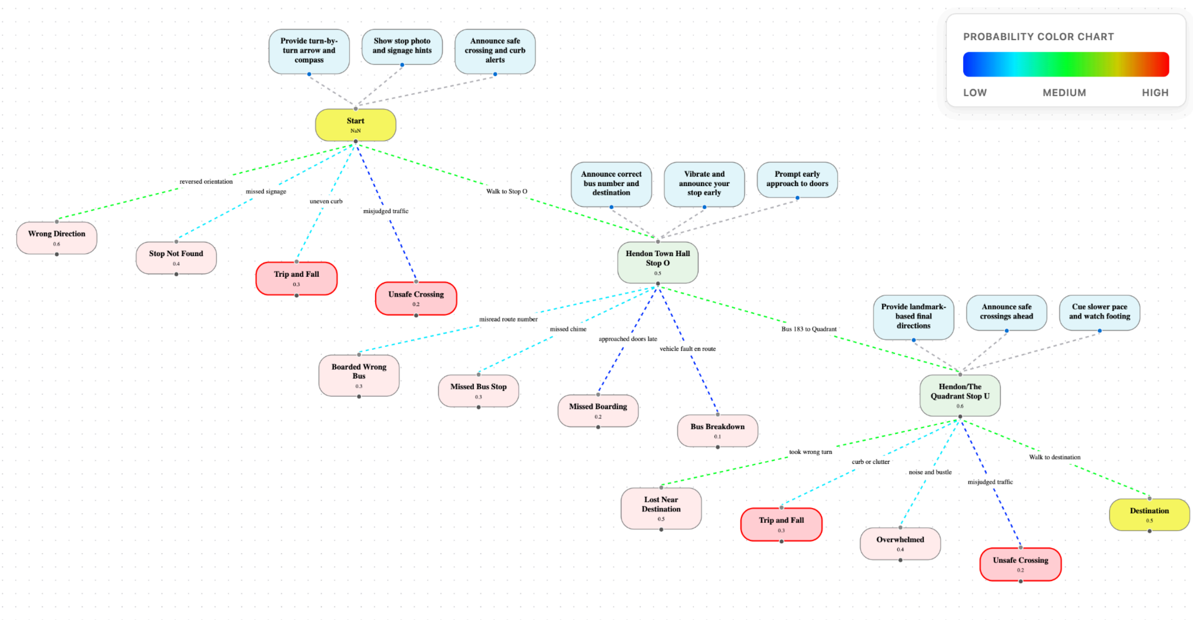
Journey graph - Frailty - Anticip8/GPT-5



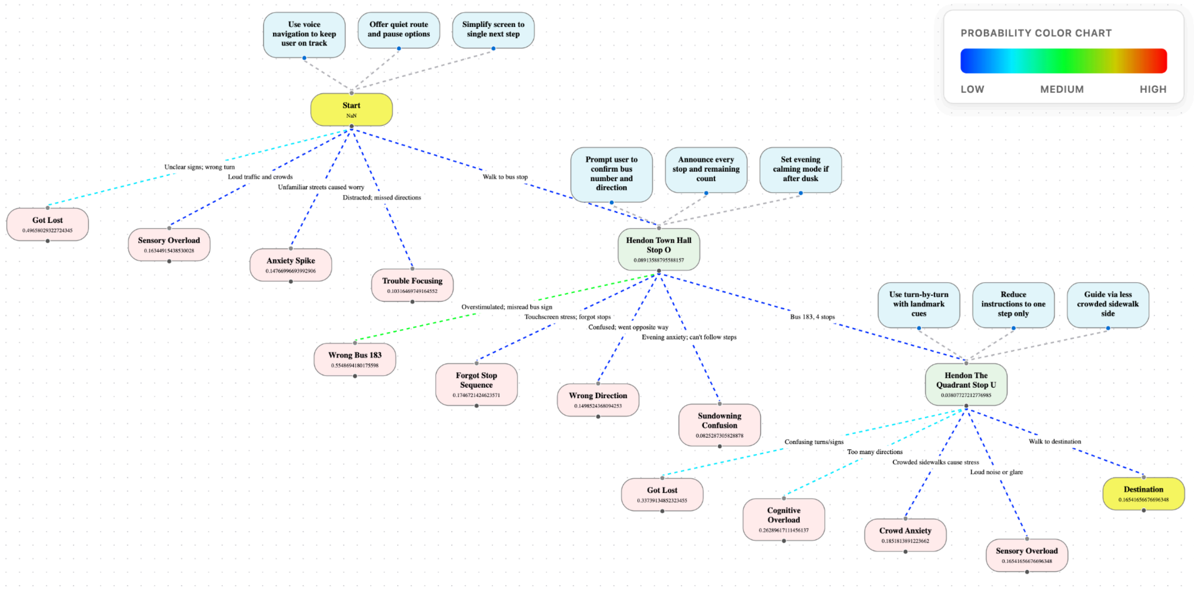
Journey graph - Dementia - GPT-5/GPT-5



Journey graph - Dementia - Anticip8/Anticip8



Journey graph - Dementia - GPT-5/Anticip8



Journey graph - Dementia - Anticip8/GPT-5

Appendix B - SafeStep User Manual

SafeStep User Manual

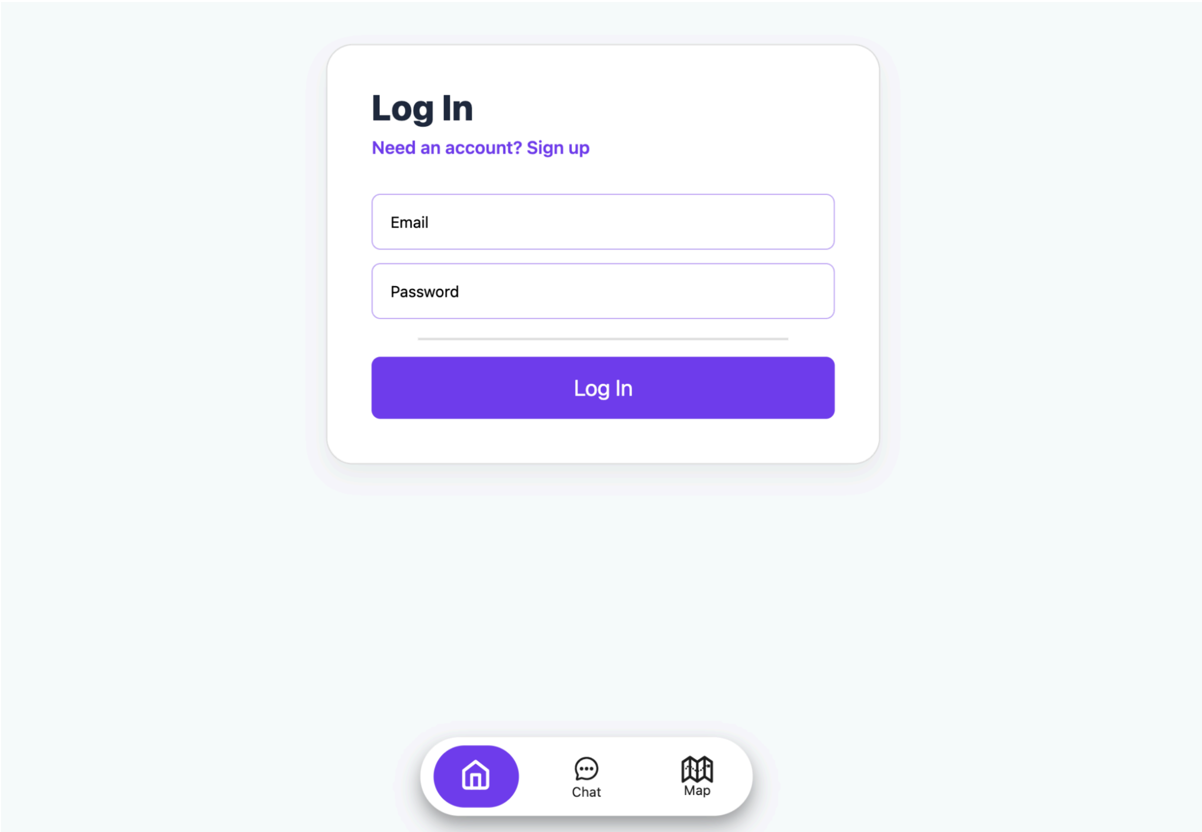
This is the user manual for using the SafeStep application, intended to test journey generation, graph and map display.

Web based instructions:

To use the web visualisation tool, users need to go to:

https://safestepcompanion.netlify.app

Once the page loads, users will be presented with a login form. Note that **users need to be logged in** to be able to use the graph generation and chat features.



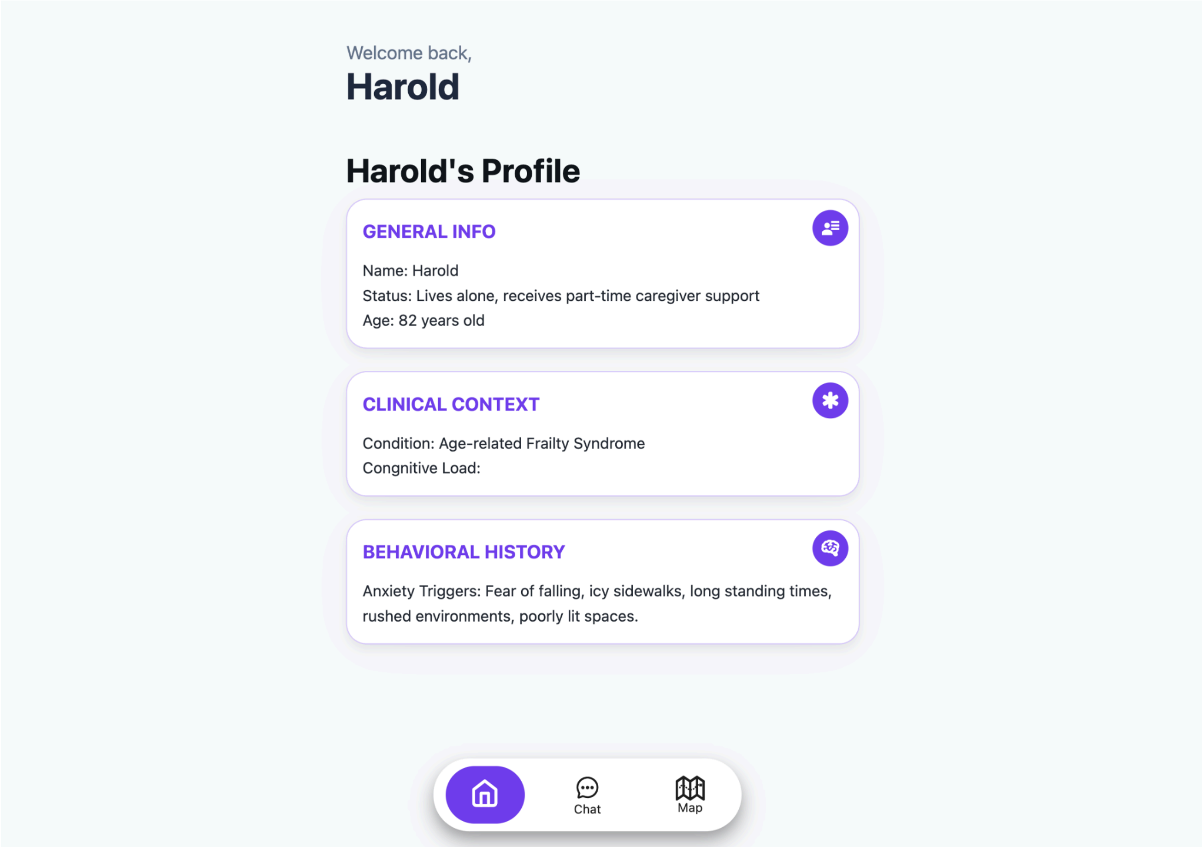
Login page

The application assigns profiles to every new user. I have prepared and handed login credentials for both Dementia and Frailty profiles.

Dementia profile:
Email: user1safestep@mail.com
Password: user1SafeStep!

Frailty profile:
Email: user4safestep@mail.com
Password: user4SafeStep!

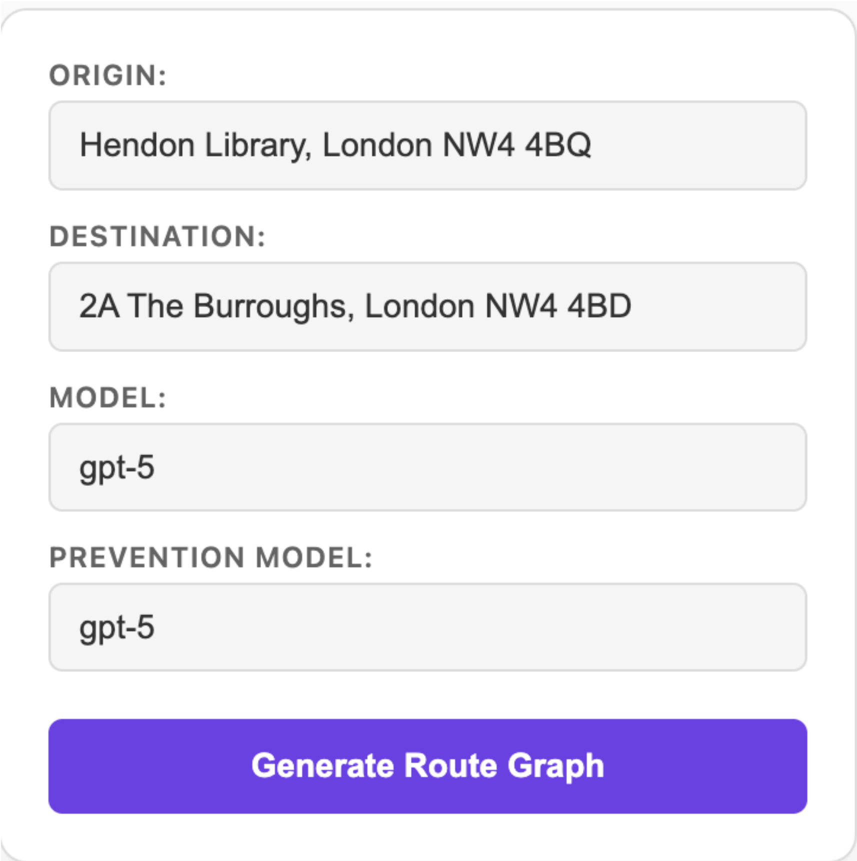
Once logged in, the current home page only displays some key insights on the user's profile, but it will eventually display a dashboard for the user and their caregiver.



Home page

While the web application is only intended to display journey graphs, journey generation through the chat feature is possible. If users want to try generating a journey graph through the chat feature, they must send a message asking to generate a route from origin to destination, like the following example: ***"Can you calculate the route from 42 Grosvenor Terrace, SE5 ONP, London to John Ruskin St, London SE5 0EN using bus as my transport mode, please?"***

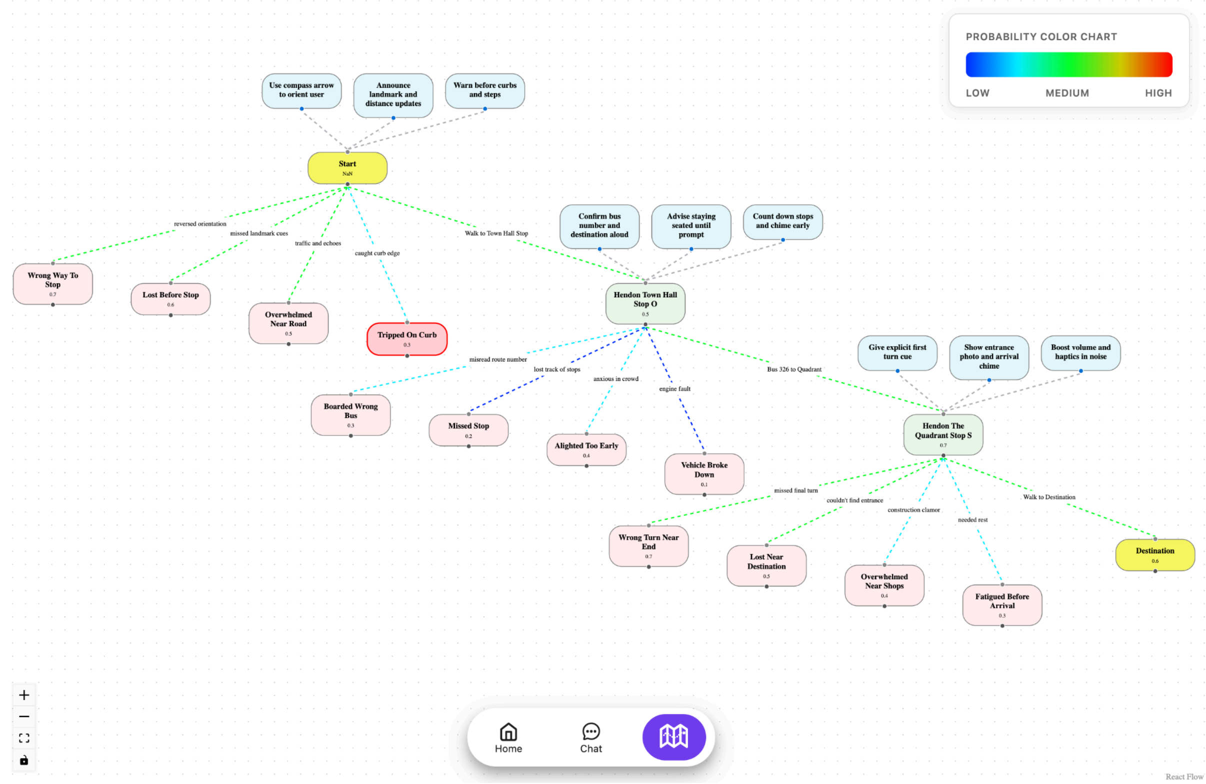
Alternatively, users can use the UI provided in the map tab to generate the journey graphs. To do so, users must fill in the Origin, Destination, Model and Model prevention inputs and click the "Generate Route Graph" button.



Journey Generation form

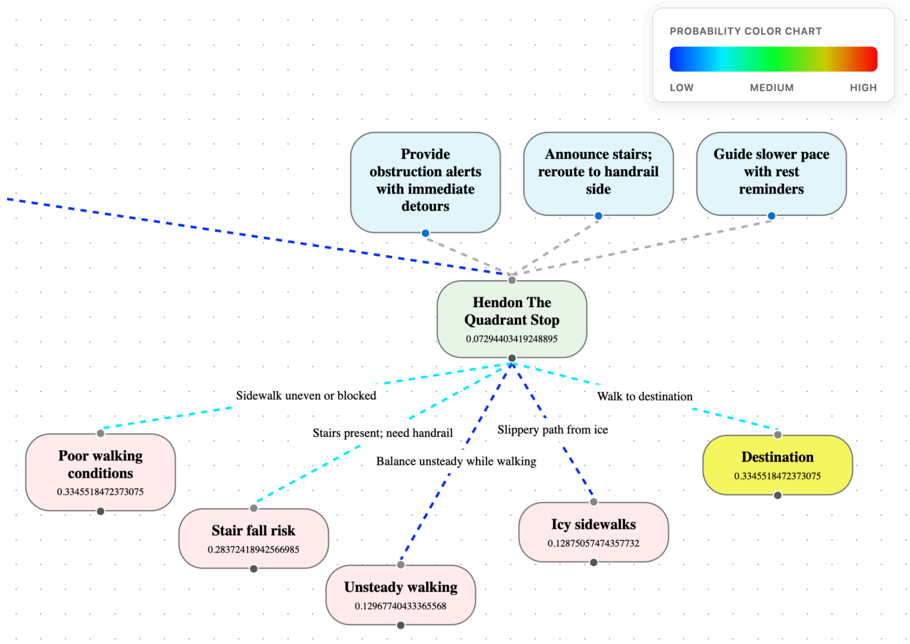
Once the Journey request is sent, there will be a loading modal for the duration of the pipeline runtime. Note that depending on the user's journey, this can take over 10 minutes, so please be patient.

Once the pipeline successfully finishes generating the journey graph, it will display it, and users will be able to interact with the generated preventions.

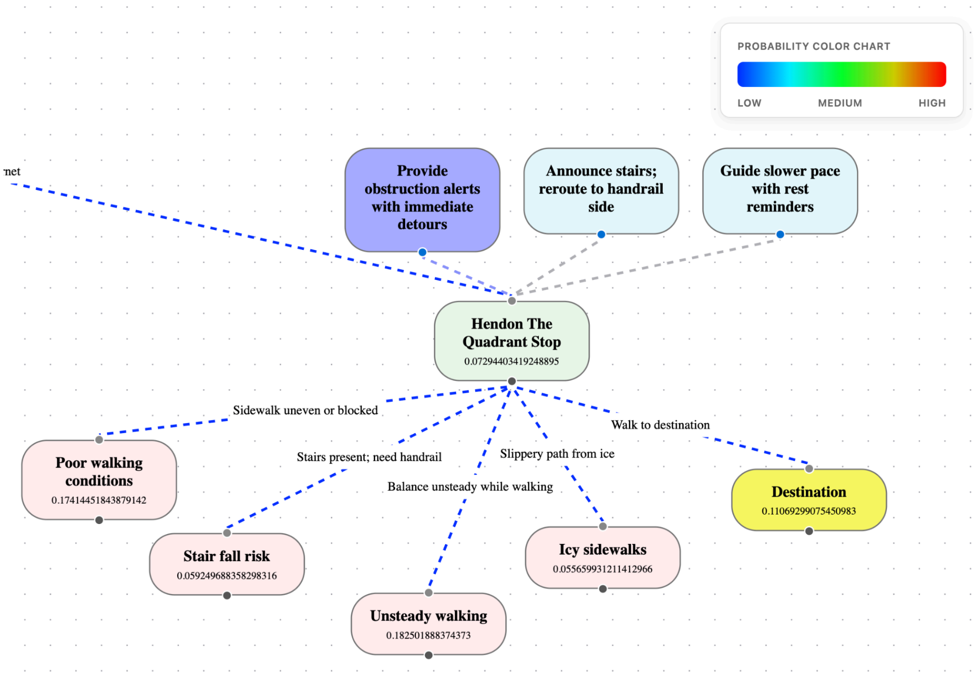


Example of a generated graph

To interact with the preventions, simply click on a specific node's prevention (the top three boxes on each node) and watch the generated probabilities for all failure nodes/intended next node change accordingly.



Node without any prevention applied.

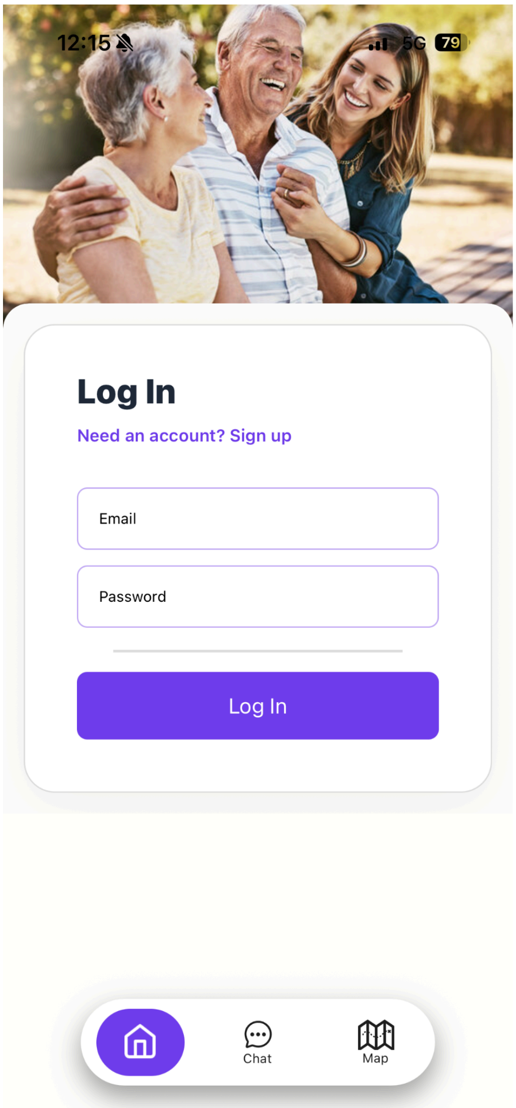


Node with first prevention applied.

If you experience any issues while using the web application, please don't hesitate to contact me about it. Your feedback is very appreciated!

Application instructions:

To use the SafeStep mobile application, please download it using the provided QR code. Users will be presented with a loading screen while the application downloads, and once it loads, it will, similar to the web application, display a login screen. Note that to use the application, users **need to be logged in** using the aforementioned credentials. Additionally, due to time constraints and ease of testing, the app has been developed to work best on an iPhone, and has been tested on iPhone X, 11, 12, 13 and 15. Therefore, full Android compatibility is not guaranteed at this stage of development.



Loading and login screen

To test the journey generation and map displays, users should move to the chat tab and prompt the system with a message asking to generate a route from origin to destination like the following example: ***"Can you please calculate the route from 42 Grosvenor Terrace, SE5 ONP, London to John Ruskin St, London SE5 0EN using bus as my transport modes please"***.

Users will be promptly replied to with an acknowledgement message while the journey generation pipeline is working. Once this process is done, users will receive a written message in the chat accompanied by a voice prompt reading the message aloud. Subsequently, users can navigate to the map tab, where they will be able to see the displayed journey steps divided into their individual polylines.

Note that a common issue with the Expo Go environment while tracking location, is the following error. This error does not interfere with the app's usage in any way, so users are free to dismiss it.

```
Console Error
Uncaught (in promise, id: 0) Error: One of the
`NSLocation*UsageDescription` keys must be present in Info.plist
to be able to use geolocation.
```

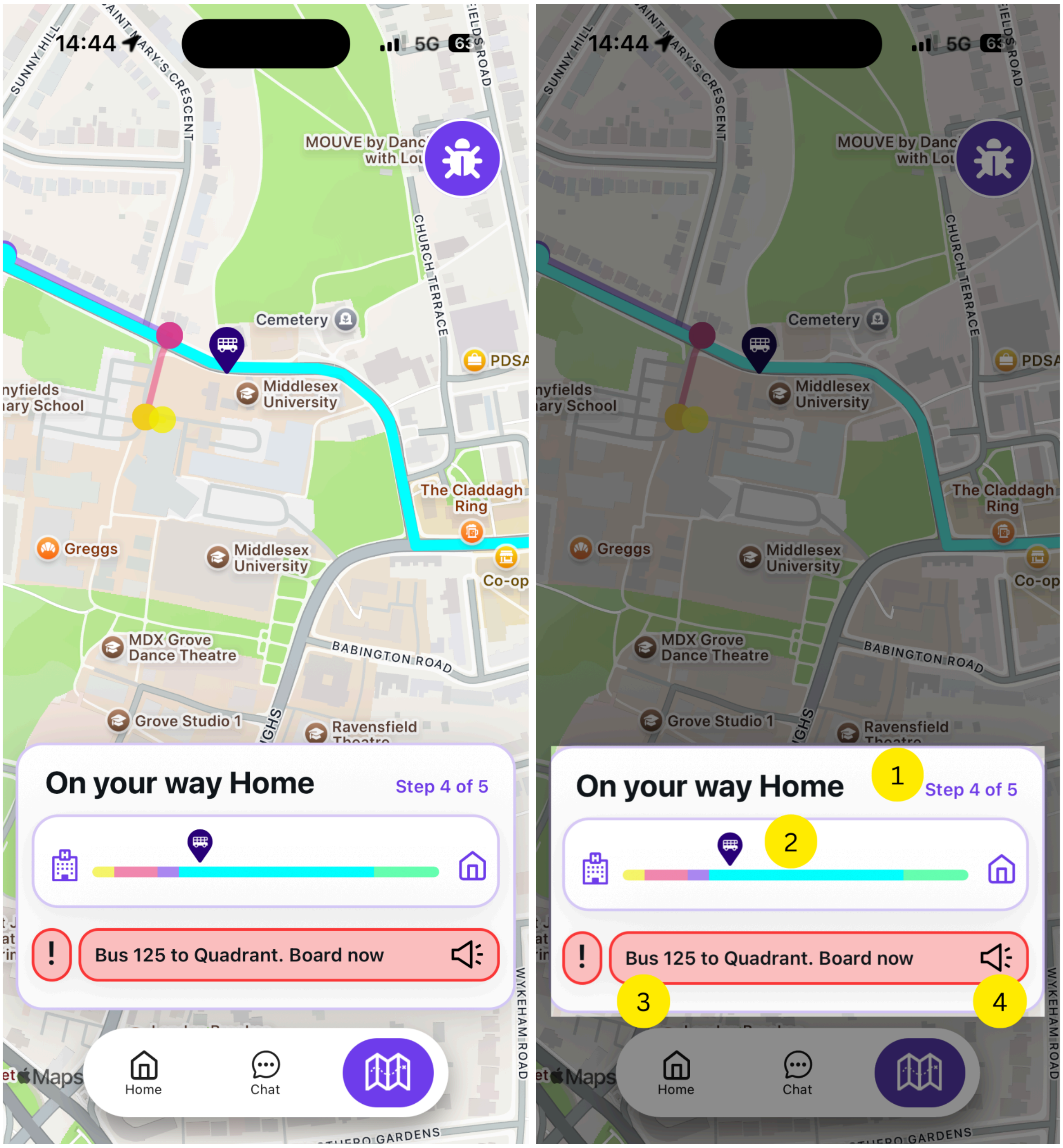
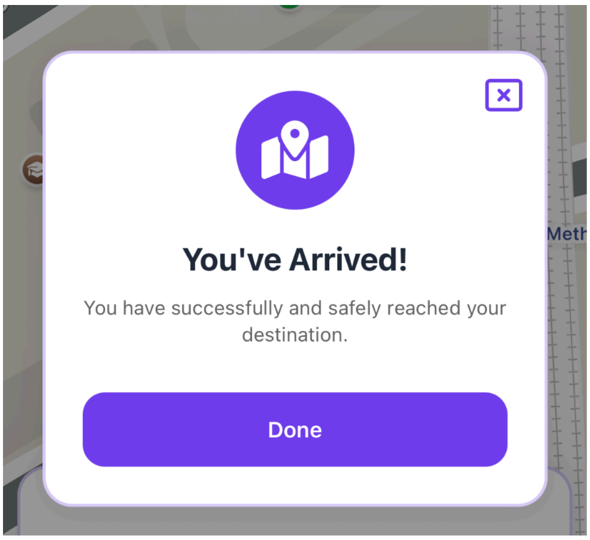


Figure x (left) and figure y (right) displaying map user interface

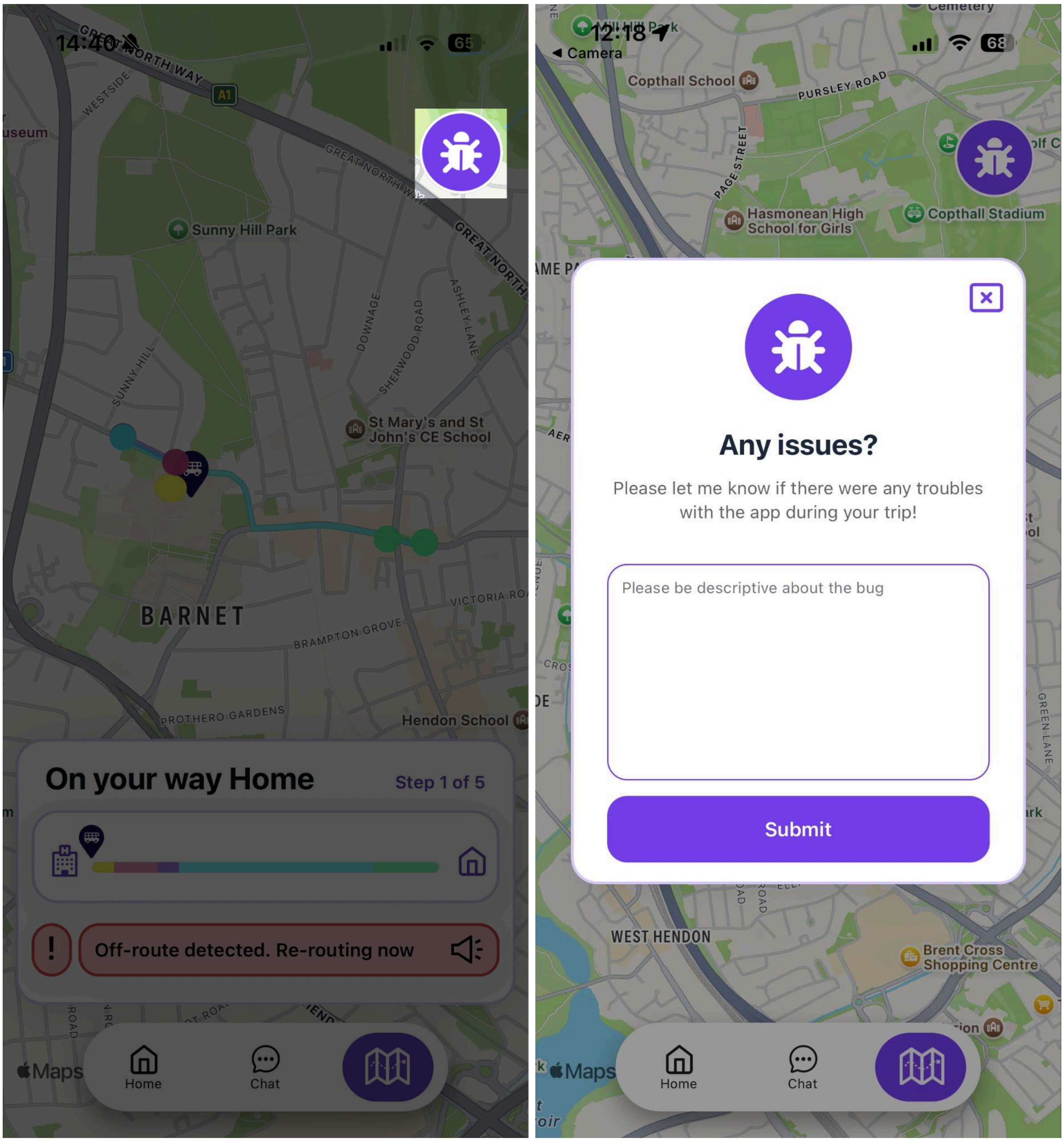
Figure x and y above show the map user interface available during any given trip. Notable information displayed by Figure x is the several individual polylines that compose a full journey, and the user's location and the highlighted current polyline representing the user's current step. Figure y, highlighting the journey information component, requires a more detailed explanation. Numbered areas represent:

- (1) The current step the user is on, compared to the total number of steps.
- (2) A line representing the whole trip, divided between the individual polylines representing each step in the journey. Additionally, a dynamic user icon is displayed to indicate the user's precise position in relation to the whole journey.
- (3) A text box intended to draw the user's attention, containing a short message crafted to increase the user's probabilities of reaching the next step.
- (4) A "speaker" button that plays a longer version of the written message, allowing users to keep informed and on track through different sources.

Once the user reaches their destination, they will be presented with a modal letting them know, after tapping the "***Done***" button, the app is free to perform another journey.



Finally, since this is an early-stage working prototype, users are encouraged to report any bugs they find, whether it's an application-breaking issue or a visual interface bug, please feel free to report it. Alternatively, users can contact me directly about any bugs or issues experienced during app testing. All feedback is appreciated.



Figures showing bug report feature

Appendix C - SafeStep Industry Write Up by Fountech AI

"Our experience at Fountech with Azul's SafeStep project – supervised by Dr. Gamez – was outstanding. The project directly tackles the UK's elderly care crisis, where frailty affects over 10% of those aged 65 and older at a cost of £5.8 billion per year to the NHS, and dementia cases are projected to reach 1.4 million by 2040 with a £90 billion burden. It empowers independent travel through London's complex TfL network.

Azul exceeded expectations by integrating our early-access Anticip8 API – our predictive behavioral AI that forecasts human actions from user profiles and context – with minimal guidance to build a robust early prototype, including FastAPI/AWS backend, React Native app, and agentic workflows anticipating risks like "Sundowning disorientation" or "stair fall risk" with calm cues and polylines. Live testing across 26 real TfL journeys from March 13-24, 2026 detected all 10 derailments in under 30 seconds, earning user praise for reassuring, relevant support – while noting font tweaks for elderly accessibility.

SafeStep showcases Anticip8's health tech potential as a proactive companion reducing isolation and NHS strain, though its ambitious scope defers features like RAG reminders, caregiver dashboards, and off-track corrections to future iterations. We're impressed by Azul's efficiency with our API, TfL data, and cloud resources within tight timelines, and deeply grateful for Dr. Gamez's excellent communication and regular coordination with our team. This prototype merits further development.

Nicole Kairinos

Product Manager, Anticip8 "